

# Scrutinized lipid utilization disrupts Amphotericin-B responsiveness in clinical isolates of *Leishmania donovani*

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## eLife Assessment

This **important** study investigates the propensity of the intravacuolar pathogen, *Leishmania*, to scavenge lipids which it utilizes for its accelerated growth within macrophages. Although some of the data compellingly links increased lipid acquisition to parasite growth, data to support the underlying mechanism to describe the proposed model is **incomplete**. The study adds to other work that has implicated pathogen-derived processes in the selective recruitment of vesicles to the pathogen-containing vacuole, based on the content of the cargo.

<https://doi.org/10.7554/eLife.102857.1.sa3>

## Abstract

The management of *Leishmania donovani* (LD), responsible for fatal visceral leishmaniasis (VL), faces increasing challenges due to rising drug-unresponsiveness, leading to increasing treatment failures. While hypolipidemia characterizes VL, LD, a cholesterol auxotroph, relies on host lipid scavenging for its intracellular survival. The aggressive pathology, in terms of increased organ parasite load, observed in hosts infected with antimony-unresponsive-LD (LD-R) as compared to their sensitive counterparts (LD-S), highlights LD-R's heightened reliance on host lipids. Here we report that LD-R-infection promotes fluid-phase endocytosis in the host, selectively accumulating neutral lipids while excluding oxidized-LDL. LD-R enhances the fusion of endocytosed LDL- vesicles with its phagolysosomal membrane and inhibits cholesterol mobilization from these vesicles by suppressing NPC-1. This provides LD-R amastigotes with excess lipids, supporting their rapid proliferation and membrane synthesis. This excess LDL-influx leads to an eventual accumulation of neutral lipid droplets around LD-R amastigotes, thereby increasing their unresponsiveness towards Amphotericin-B, a second-line amphiphilic antileishmanial. Notably, VL patients showing relapse with Amphotericin-B treatment exhibited significantly lower serum LDL and cholesterol than cured cases. Treatment with Aspirin, a lipid droplet blocker, reduced lipid droplets around LD-R amastigotes, restoring Amphotericin-B responsiveness.

## Introduction

Antimony, which was mainstay for treating fatal visceral leishmaniasis (VL), has been withdrawn from Indian Subcontinent for more than a decade now due to emergence of resistance. Despite this, recent clinical LD isolates remain unresponsive to antimony (LD-R), indicating persistent phenotype [1]. Notably, LD-R also exhibits more aggressive pathology in clinical and experimental infection featuring increased organ parasite burden as compared to LD-S-infection [2–5]. Increased metacyclogenesis among LD-R was initially thought to be the primary factor contributing to higher infectivity, resulting in organ-parasite overload [6]. However, later studies revealed that LD-R promastigotes have a greater replicative potential than LD-S, possibly offering a selective intracellular survival advantage [7, 8]. This implies LD-R might be more adept at acquiring and metabolizing host derived nutrients within their parasitophorous vacuole (PV), facilitating their heightened intracellular proliferation. Lipids are crucial carbon sources for energy in proliferating intracellular pathogens. Several intracellular bacteria, protozoans and viruses utilizes cellular lipid droplets and plasma lipoproteins for entry, energy acquisition and biomembrane synthesis, demonstrating intricate interactions between intracellular pathogens and host lipid metabolism [9, 10]. *Leishmania* parasites are cholesterol auxotrophs [11], and need (i) host membrane cholesterol for successful attachment and infection [12]. (ii) Intracellular leishmania amastigotes replicating within membrane-bound PVs, accumulates cholesterol and lipid-droplets [13]. (iii) *Leishmania* amastigotes isolated from infected host have been shown to have increased cholesterol content in them [14], all these clearly suggest replicating *Leishmania* amastigotes might co-opt host derived lipids to facilitate their intracellular proliferation and membrane synthesis. In fact, hypocholesteremia is a hallmark of VL, with intracellular LD depleting host cholesterol and lipoproteins [15, 16]. However, the mechanism by which PV-bound LD amastigotes acquire this lipids, and how variations in lipid type, source, and utilization dynamics affect disease pathology and drug responsiveness, remains unclear.

So far, drug responsiveness in leishmaniasis has mostly been studied through in vitro selection of drug-resistant promastigotes, with limited focus on the drug susceptibility profiles of circulating clinical strains of *Leishmania* [17–19]. However, growing evidences indicates that drug responsiveness and treatment failure in VL are multifactorial, influenced by the host's immune-metabolic status [3, 20]. Importantly, recent reports have also linked emergence of drug unresponsiveness among intracellular pathogens with their ability to modulate host lipid metabolism [21, 22]. For instance, increased lipid accumulation can hinder drug penetration by trapping lipophilic drugs, fostering resistance [23–25]. VL treatment in recent years have seen a rise in treatment failure against Amphotericin-B (Amp-B), a second-line lipophilic drug [2], making it critical to investigate these failures in the context of lipid utilization during LD infection.

Using clinical antimony unresponsive LD-strains isolated from patients this study elucidates how LD-R-infection leads to an increased uptake of low-density lipoprotein (LDL) leading to eventual accumulation of neutral lipids droplets within their PV. In vitro infection of liver-resident Kupffer cells (KCs), primary sites for cholesterol synthesis and LD amastigote residence, demonstrated that LD-R infection facilitates fluid-phase endocytosis, increasing LDL-vesicle uptake in infected-KCs cells. LD-R-PV exhibits a higher propensity to fuse with endocytosed LDL-vesicles, coupled with the inhibition of cholesterol recycling from the vesicles back to the plasma membrane of Infected-MFs. Our observation thus explains the heightened energy source and cholesterol availability for PV and daughter membrane synthesis to accommodate higher number of LD-R amastigotes observed in clinical infection. We also observed a suppressed expression of MSR-1, a scavenger receptor of ox-LDL [26], in LD-R infected murine liver, suggesting LD-R-infection limits the uptake of ox-LDL, reducing inflammatory responses. Furthermore, our findings indicate that lipid droplet

accumulation around LD-R amastigotes contributes to Amp-B unresponsiveness, explaining recent reports of Amp-B cross-resistance in clinical LD-R isolates from India [27, 28]. Significant low lipid profiles in VL patients experiencing treatment failure against liposomal Amp-B, combined with complete clearance of LD-R amastigotes using a combination treatment of Aspirin (a lipid droplet formation blocker) and Amp-B, confirmed critical role of assimilating host lipids in developing Amp-B unresponsiveness among rapidly replicating LD amastigotes exhibiting primary unresponsiveness towards antimony. This study identifies sequence of events that enables LD-R amastigotes to redirect host nutrients for their own aggravated sustenance, and how this preferential lipid utilization might equip them with unresponsiveness against Amp-B contributing to the gradual spread of cross-resistance among clinical LD-R isolates.

## Results

### 1. LD-R infection caused severe dyslipidaemia as compared to LD-S infection

Intracellular amastigotes of *Leishmania* parasites exhibit lipid auxotrophy [11], with dyslipidaemia being a hallmark for leishmaniasis [15], including VL patients. Previously, LD amastigote load in splenic aspirates of VL patients have been shown to exhibit an inverse correlation with serum cholesterol level [29]. Since infection with antimony-unresponsive-LD (LD-R) has been previously reported to exhibit higher amastigote load in spleen and liver of infected host, as compared to infection with their responsive counterparts (LD-S) [7, 30], we wanted to investigate possible extent of dyslipidaemia in response to LD-R and LD-S-infection and how it might affect disease pathology. Mice infected with equal number of sorted metacyclics (**Fig S1A**) of natural clinical LD-isolates with confirmed LD-R phenotype, MHOM/IN/09/BHU575/0 (LD-R<sup>1</sup>) and MHOM/IN/10/BHU814/1 (LD-R<sup>2</sup>), showed a significantly lower serum levels of LDL, HDL, and cholesterol as compared to MHOM/IN/83/AG83 (LD-S<sup>1</sup>) and MHOM/IN/80/DD8 (LD-S<sup>2</sup>), infected mice (**Figure 1A**). Liver serves as a primary site for lipid synthesis and liver-resident MFs (Kupffer cells, KC) are one of the primary sites harboring LD amastigotes [31]. Therefore, we decided to test if KCs can be established as an *in vitro* model for LD-infection to investigate the possible link of host lipid utilization with respect to aggressive pathology shown by LD-R. Purified KCs (**Figure S1B**) infected with equal number of sorted metacyclics for LD-S and LD-R-isolates as before displayed no significant difference in their initial attachment to host membrane (**Figure 1B** and **Movie 1A, 1B**). However, although initial number of intracellular LD-S and LD-R-amastigotes 4Hrs post infection (p.i) showed no difference, a significantly higher number of LD-R amastigotes were observed in 24Hrs infected-KC (**Figure 1Ci, ii**). Parallel infection performed in peritoneal exudate macrophages (PEC), which serves as a long-standing model for *in vitro* LD-infection, under identical experimental conditions also resulted in similar observation with LD-R strains showing significantly higher amastigote load at 24Hrs p.i than LD-S strains (**Figure S1Ci, ii, iii**), as previously reported by multiple groups [1, 4, 6]. Moreover, KCs infected with LD-S or LD-R amastigotes isolated from 28 days infected mouse spleens, also resulted in significantly higher parasite load for LD-R (**Figure S1D**), clearly indicating that higher parasite load is a conserved LD-R phenotype which does not depend on initial infectivity or MF origin. Notably, infected-KCs had a slightly higher inherent amastigote load compared to infected-PECs for both LD-S and LD-R-infections, with no significant difference in parasite load within the two independent LD-S or LD-R strains (**Figure 1C**, **Figure S1C, S1D**). Hence, BHU575 was selected as representative LD-R and AG83 as representative LD-S to conduct all further comparisons with the inclusion of other representative strains as and when mentioned. Higher intracellular amastigote load in response to LD-R-infection was further confirmed by imaging LD-S and LD-R-PV at 4Hrs and 24Hrs infected-KCs (**Figure 1D**). Comparative analysis revealed significantly higher number of LD-R-PVs as compared to LD-S-PVs at 24Hrs p.i, although this difference was not

observed at 4Hrs p.i (**Figure 1Di, Dii**). Interestingly, 3D image reconstruction for LD-R-PV at 24Hrs p.i, presented it as a larger fused structure as compared to smaller individual segmented appearance of LD-S-PV (**Figure 1Diii**).

## 2. Lipid deprivation is detrimental for LD-R while inherent hypercholesterolemia facilitates its proliferation

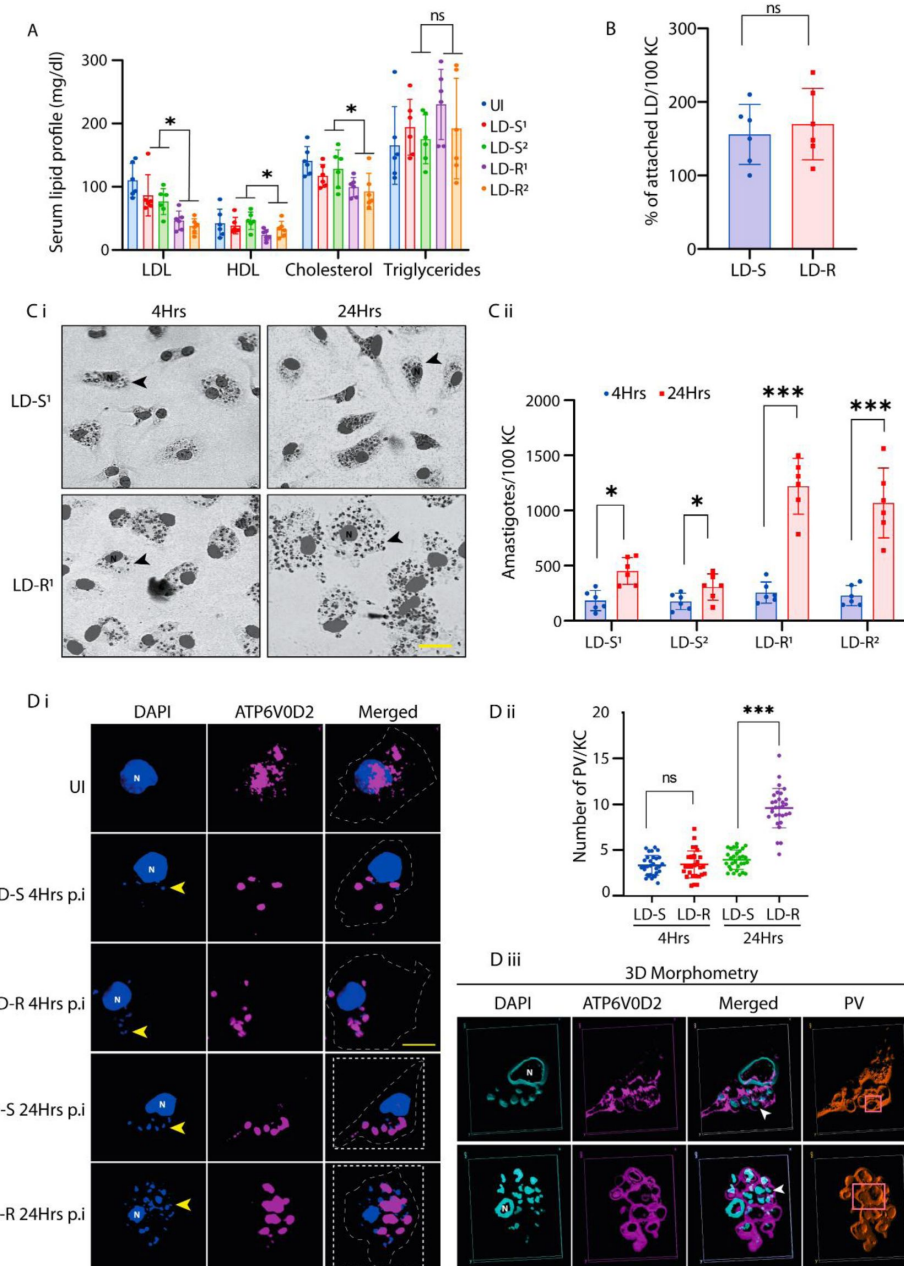
To determine if extracellular lipids are absolutely essential for proliferation of LD-R amastigotes, LD-S and LD-R-infections were performed in high, low or lipid-free medium. Reduction in extracellular lipids severely compromised proliferation of LD-R but not LD-S amastigotes in infected-KCs, confirming LD-R's dependency on external lipid availability (**Figure 2A i, ii**). Alternatively, infection performed in KCs isolated from ApoE-knockout (KO)-mice(*ApoE*<sup>-/-</sup>) with inherent hyperlipidemia [32], resulted in significant increase in amastigote load as compared to wild type BL/6-infected-KCs, only for LD-R-infection, with LD-S infection resulting in comparable amastigote load (**Figure 2B i, ii**). Previously, LD-S-infections in *ApoE*<sup>-/-</sup> mice have been reported to typically get cleared by 42 days p.i, due to heightened T-cell activity [20]. Interestingly, however, we observed *ApoE*<sup>-/-</sup> mice infected with GFP expressing LD-S or LD-R resulted in increased splenic parasite burden as compared to wild type BL/6 mice at an earlier time point, 11 days p.i (**Figure 2C i, ii**). This observation possibly indicates a higher initial infection propensity in *ApoE*<sup>-/-</sup> mice, which might get resolved later on due to increasing T-cell activity. Our observation also showed that splenic parasite load for LD-R is significantly higher in *ApoE*<sup>-/-</sup> than BL/6 (**Figure 2C ii, iii**) again confirming that higher availability of extracellular lipids might help in increased proliferation of LD-R amastigotes.

## 3. Quenching of host membrane cholesterol remains consistent during initial infection with LD-S and LD-R

Host lipids scavenging is essential for any intracellular pathogens exhibiting lipid auxotrophy [22]. Thus, to support their aggressive division, LD-R amastigotes should require significantly higher amount of host lipids, which in turn will allow excess supply of cholesterol for increased PV and daughter membrane biogenesis as compared to LD-S amastigotes. *Leishmania* amastigotes are lipid auxotrophs [11, 14], and it has been previously reported that *L. Mexicana* can acquire host membrane cholesterol during its initial entry into host MFs [33]. Thus, host membrane Cholesterol was investigated as a possible source and potential mechanism by which LD-R acquires host-derived lipids distinctively from their LD-S counterparts. However, confocal image analysis of NBD-Cholesterol labeled KCs infected with RFP expressing LD-S or LD-R revealed that both LD-S and LD-R can extract NBD-Cholesterol from the KC membrane during their initial infection (4Hrs p.i) with comparable efficiency (**Figure S2A i, ii**). This observation was further confirmed by quantification of NBD-Cholesterol between freshly invaded LD-S and LD-R parasites isolated from infected-KC by flow cytometry (**Figure S2B i, ii**).

## 4. LD-R-infection mobilizes extracellular LDL within infected-KCs to support its intracellular proliferation

Since the quenching of host membrane cholesterol during initial entry of LD promastigotes does not significantly differ in LD-S and LD-R mediated infections (**Figure S2**), we explored other potential host lipid sources for supporting LD-R's increased intracellular division and membrane synthesis. LDL serves as the primary source of cellular lipids which helps to maintain lipid homeostasis, and gets readily metabolized to produce bioavailable cholesterol [34]. Our previous observation showing a significant drop in LDL levels between LD-S and LD-R-infected mice (**Figure 1**), led us to investigate LDL as a possible source which can differentially contribute in providing excess lipids, sterols, and fatty acids to rapidly proliferating intracellular LD-R amastigotes. Live cell imaging comparing uptake of fluorescent red-LDL in EGFP-LD-S or

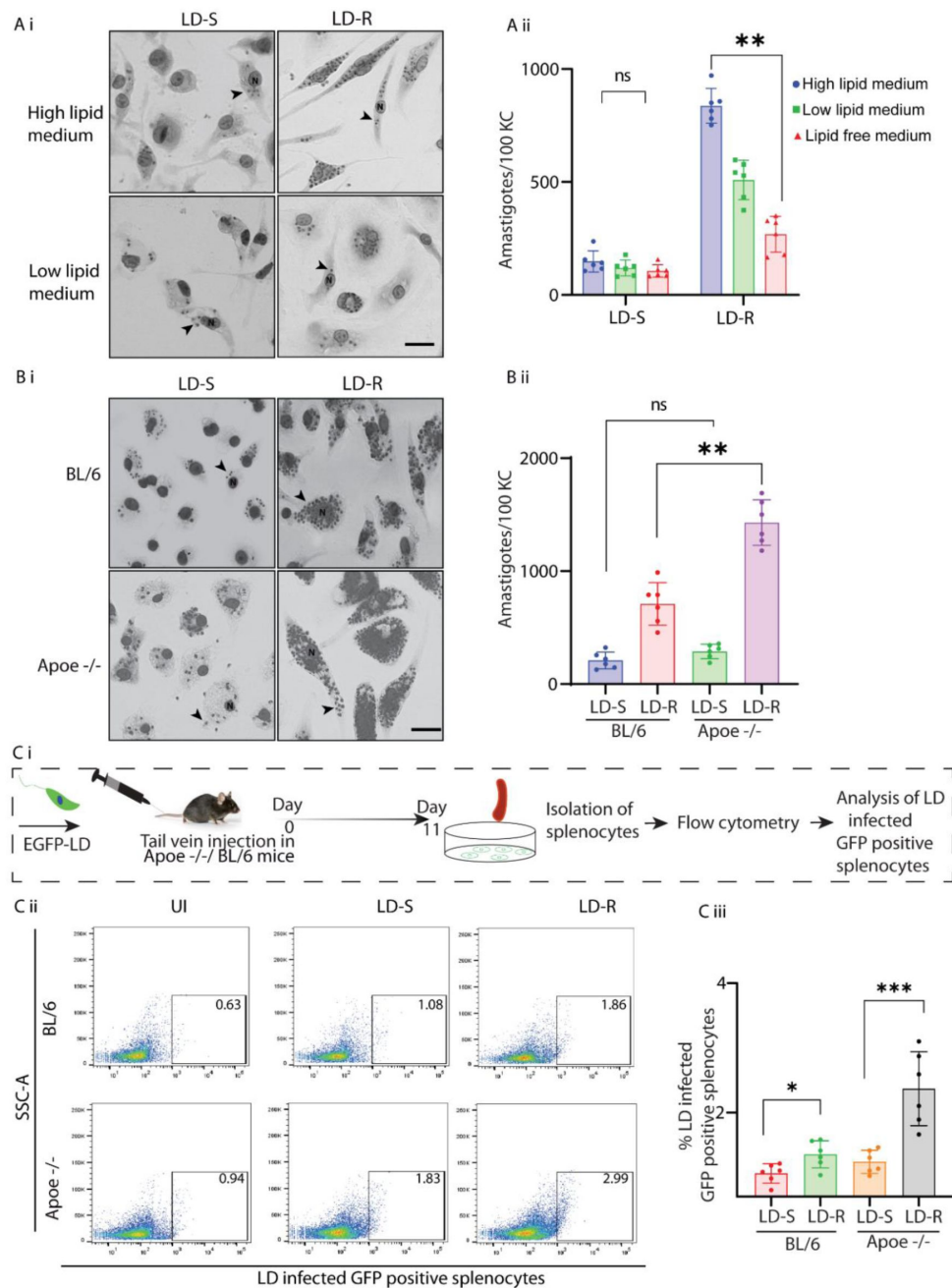


**Figure 1.**

### Severe dyslipidemia is linked with higher intracellular proliferation of LD-R amastigotes.

**(A)** Serum lipid profile of 28 days LD-infected Balb/c mice (N=5). Equal number of sorted metacyclic promastigotes of two representative LD-S (LD-S<sup>1</sup>, LD-S<sup>2</sup>) and LD-R (LD-R<sup>1</sup>, LD-R<sup>2</sup>) strains were used to perform independent infection. **(B)** Number of attached LDs on KC surface 4Hrs p.i, data represents (N=3) for two LD-S and two LD-R strains. **C (i)** Representative Giemsa stain images of LD-S<sup>1</sup> and LD-R<sup>1</sup>-infected KC at 4Hrs and 24Hrs p.i. Black arrow represents LD amastigotes. N denotes host nucleus. Scale bar 20µm. Giemsa Images are represented in grey scale to clearly represent LD nucleus (black arrow). **C (ii)** Intracellular amastigote count for 4Hrs and 24Hrs p.i. Each dot represent count from 100 infected-KC (N=6). **D (i)** LD-infected-KC imaged to visualize the PV at 4Hrs and 24Hrs p.i. PV marked by ATP6V0D2 (magenta), KC and LD nucleus (blue), Scale bar 5µm. Yellow arrow represents LD amastigotes. In the merged image, white dotted line marks KC's periphery, and host nucleus is represented with N. **D (ii)** PV counts measured from 30 infected KCs. **D (iii)** 3D morphometric representation for better visualization of PV in 24Hrs post infected KC marked with white dotted box as in D (i). ATP6V0D2 (magenta), nucleus (cyan) with last panel showing PV-grooves marked in red solid line with white arrow representing LD amastigotes. \*\*\* signifies p value < 0.0001, \* signifies p value < 0.05, n.s non-significant.





**Figure 2.**

### LD-R-infection results in a higher organ parasite load in hypercholesterolemic *Apoe*<sup>-/-</sup> mice. **A**

**(i)** Representative Giemsa images of LD-infected-KCs cultured in high and low lipid media. Black arrow represents LD amastigotes; N represents host nucleus. Scale bar 20μM. **A (ii)** Graph representing number of intracellular LD amastigotes in KCs with LD infection performed either in high, low or lipid free media (N=6). **B (i)** Giemsa images of LD-infected-KCs isolated from wild type BL/6 and *Apoe*<sup>-/-</sup> mice. Scale bar 20μm, N representing host nucleus. Giemsa Images are represented in grey scale to clearly represent LD nucleus (black arrow). **B (ii)** Amastigote count from 100 LD-infected-KCs (N=6) as in B(i). Data are presented as mean ± SEM. **C (i)** Schematic representation of *in-vivo* LD-infection in BL/6 and *Apoe*<sup>-/-</sup> mice performed with EGFP-LD-S or EGFP-LD-R **C (ii)** Flow cytometry representing GFP-positive splenocytes isolated from 11 days EGFP-LD infected BL/6 and *Apoe*<sup>-/-</sup> mice. The black box indicates % of GFP-positive splenocytes (LD-infected). **C (iii)** Graph representing data from six infected mice as in C (ii) are presented. \*\*\* signifies p value < 0.0001, \*\* signifies p value < 0.001, \* signifies p value < 0.05, n.s non-significant.

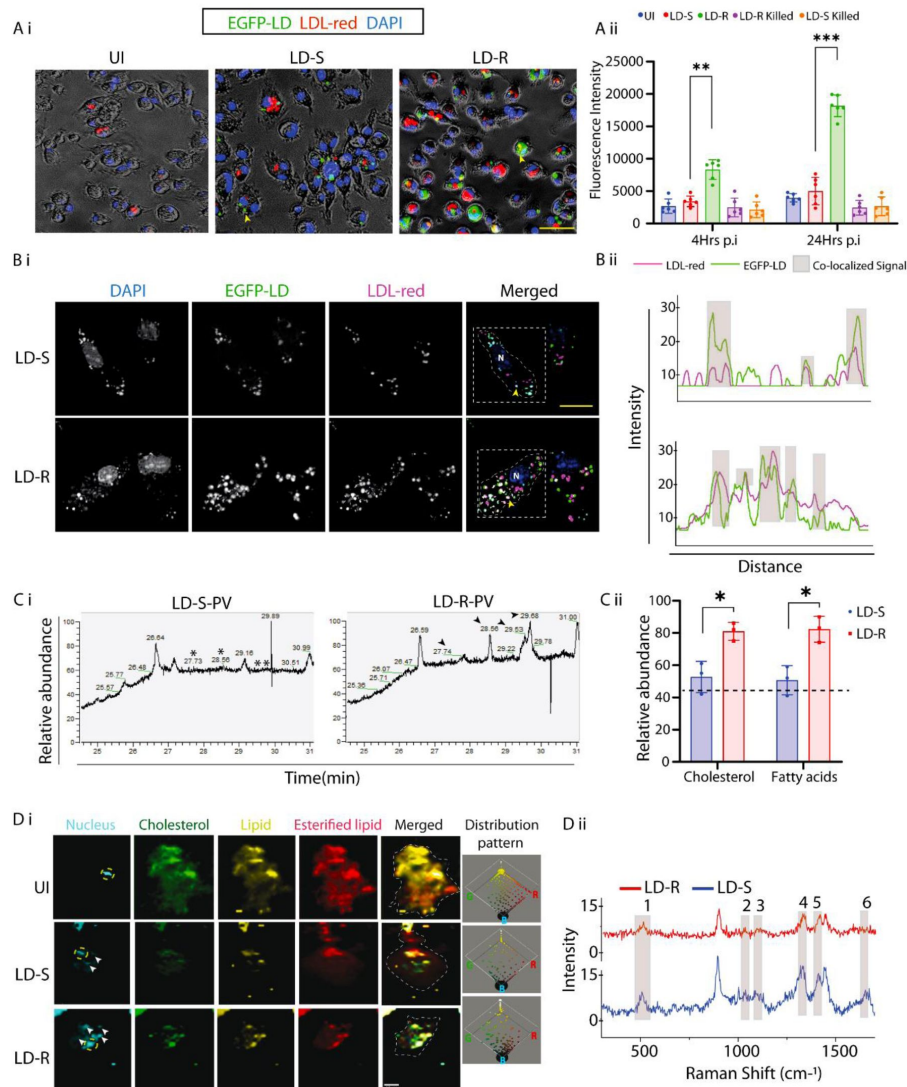
EGFP-LD-R-infected KCs at 24Hrs p.i revealed a significantly higher red-LDL accumulation within EGFP-positive LD-R- infected KCs as compared to LD-S-infection (**Figure 3Ai** [\[35\]](#)). Fluorescence plate reader-based quantification further supported this observation showing significantly higher LDL accumulation in LD-R-infected-KCs over time, which was not observed by performing infection with killed parasites (**Figure 3A ii** [\[35\]](#)), indicating that, LDL-influx is mediated by replicating LD amastigotes. Confocal microscopic images at 24Hrs p.i. of LD-S and LD-R-infected-KCs confirmed significantly higher LDL co-localizing with LD-R amastigotes compared to LD-S amastigotes (**Figure 3B i, ii** [\[35\]](#)).

## 5. LD-R-infection results in a distinct metabolic shift in lipid peaks of infected host

Next, we wanted to confirm if higher LDL-uptake could result in accumulation of excess lipids, particularly cholesterol, in LD-R-PV that would provide LD-R amastigotes with higher replicative capabilities and associated enhanced membrane synthesis potential. PV fractions were purified from LD-S and LD-R-infected-KCs post 24Hrs infection, as reported previously [\[35\]](#). Quantitative analysis of total cholesterol using Amplex red kit (**Figure S3A** [\[35\]](#)) and thin layer chromatography (TLC) (**Figure S3B** [\[35\]](#)) revealed significantly higher cholesterol content in LD-R-PV as compared to LD-S- PV. Furthermore, GC-MS of LD-S and LD-R-PV under identical experimental condition identified prominent peaks for cholesterol and fatty acids in LD-R-PV which were missing for LD-S- PV (**Figure 3C i, ii** [\[35\]](#)). Apart from this biochemical characterization, we also performed biophysical analysis of lipid related peaks using non-invasive Confocal Raman spectroscopy to capture a snapshot of lipid distribution in LD-S and LD-R-infected-KCs under native conditions (**Figure 3D i, ii** [\[35\]](#)). This analysis highlighted a distinct clustering of lipid peaks representing cholesterol, triglycerides, and fatty acids [\[36\]](#) around LD-R amastigotes, contrasting with their minimal clustering around LD-S amastigotes which resemble uninfected-KCs more closely (**Figure 3D i** [\[35\]](#)). Raman Shift analysis within the biologically active region ( $300\text{cm}^{-1}$ - $1800\text{cm}^{-1}$ ) [\[37\]](#), further confirmed a significant difference in relative abundance for lipid related peaks' at wavelengths:  $540\text{--}560\text{ cm}^{-1}$  (cholesterol),  $1080\text{--}1090\text{ cm}^{-1}$  (phospholipids),  $1270\text{--}1280\text{ cm}^{-1}$  (triglycerides),  $1300\text{--}1340\text{ cm}^{-1}$  (amide-III bond),  $1440\text{--}1453\text{ cm}^{-1}$  (fatty acids and triglycerides) [\[36\]](#), along with olefinic stretch of C=C double bonds at  $1,655\text{ cm}^{-1}$  between LD-S and LD-R-infected-KC, at 24Hrs p.i. (**Figure 3D ii** [\[35\]](#) and **S3C i, ii** [\[35\]](#)). We also noticed a distinct shift by plotting area under the curve from (Right to Left) in response to LD-R-infection (**Figure S3 D** [\[35\]](#)), indicative of distinctly different lipidomes [\[39\]](#).

## 6. LDL-uptake in LD-infected-MFs occurs through fluid phase endocytosis

As LD-R amastigotes heavy reliance on increased host lipid acquisition for rapid proliferation became evident, we wanted to investigate molecular basis of this acquisition. Considering that MFs primarily uptake LDL via the LDL receptor (LDLr) [\[40\]](#), we checked its status in response to LD-S and LD-R-infection. However, immunofluorescence analysis showed a decrease in LDLr expression in infected-KCs harboring LD-amastigotes as compared to adjacent uninfected KCs, regardless of infection type (**Figure 4A** [\[35\]](#)), along with no significant changes in total LDLr expression between uninfected, LD-S, and LD-R-infected-KCs, as observed from Western blot (**Figure S4A** [\[35\]](#)). This indicated that LDL-uptake by LD-infected-MFs might be receptor-independent. Our attempts to generate siRNA mediated Knock down (KD) in KCs were unsuccessful, hence to convincingly determine the role of LDLr in LD-infected-MFs we generated LDLr-KD-PEC (**Figure S4B i** [\[35\]](#)) and performed infection with LD-S or LD-R. Infecting LDLr-KD-PEC with LD-S or LD-R resulted in no significant changes in intracellular amastigote numbers for both infection types as compared to scrambled RNA control (**Figure S4B ii** [\[35\]](#)) confirming that although LD-R relies on increased LDL- influx for heightened proliferation, this influx is LDLr-independent. Apart from LDLr, MFs can also acquire LDL through receptor-independent fluid phase endocytosis



**Figure 3.**

### Endocytosed-LDL is the primary lipid source for intracellular LD-R proliferation.

**A (i)** Live cell images of LDL-endocytosis in 24Hrs post infected KCs with LDL (red), LD (green), and the nucleus (blue). Images were acquired directly from 96 well glass bottom plate and bright field was merged with reduced brightness and increased contrast identically for all experimental conditions. Scale bar 40µm, Yellow arrow represents intracellular LD. **A (ii)** Graph representing Fluorescence plate reader-based quantification of A (i). **B (i)** Co-localization of LDL with the LD- amastigotes in infected KCs 24Hrs p.i. Merged image showing LDL (magenta), LD (green), nucleus (blue), White dotted line in the merged image demarcates infected-KC for co-localization analysis. Scale bar 10µm, yellow arrow represents LD nucleus and N represents host nucleus **B (ii)** Intensity plot representing co-localized signals from B (i), marked with grey transparent box. **C (i)** GC-MS lipid profile of isolated PVs from LD-infected-KCs. Retention time 27.73 corresponding to the cholesterol and 28.56 corresponding to fatty acids are represented with \* for LD-S-PV and with black arrowheads for LD-R-PV. **C (ii)** Relative abundance of Cholesterol and Fatty acids of C (i). **D (i)** Representative Confocal Raman spectroscopy image of uninfected or LD-infected-KCs 24Hrs p.i, illustrating distinct lipid-related signal distribution patterns marked with pseudo-colours. Yellow dotted circle in the left most panel demarcates the host cell nucleus while white arrow marking intracellular LD nucleus. In merged panel white dotted line demarcates cells periphery. The right most panel shows dot plot representation of lipid distribution, where each colour corresponds to different lipid-related peaks with respect to blue (B) indicating the nucleus. **D (ii)** Comparative Raman spectra from LD-S and LD-R-infected-KCs. Lipid-related peaks are demarcated with shaded box, representing 1. 540-560 cm<sup>-1</sup> (Cholesterol), 2. 1080-1090 cm<sup>-1</sup> (phospholipids), 3. 1270-1280 cm<sup>-1</sup> (triglycerides), 4. 1300-1340 cm<sup>-1</sup> (Amide-III bond), 5. 1440-1453 cm<sup>-1</sup> (Fatty acids and Triglycerides), and 6. 1650-1660 cm<sup>-1</sup> (Amide-I bond). \*\*\* signifies p value < 0.0001, \*\* signifies p value < 0.001, \* signifies p value < 0.05, n.s non-significant.



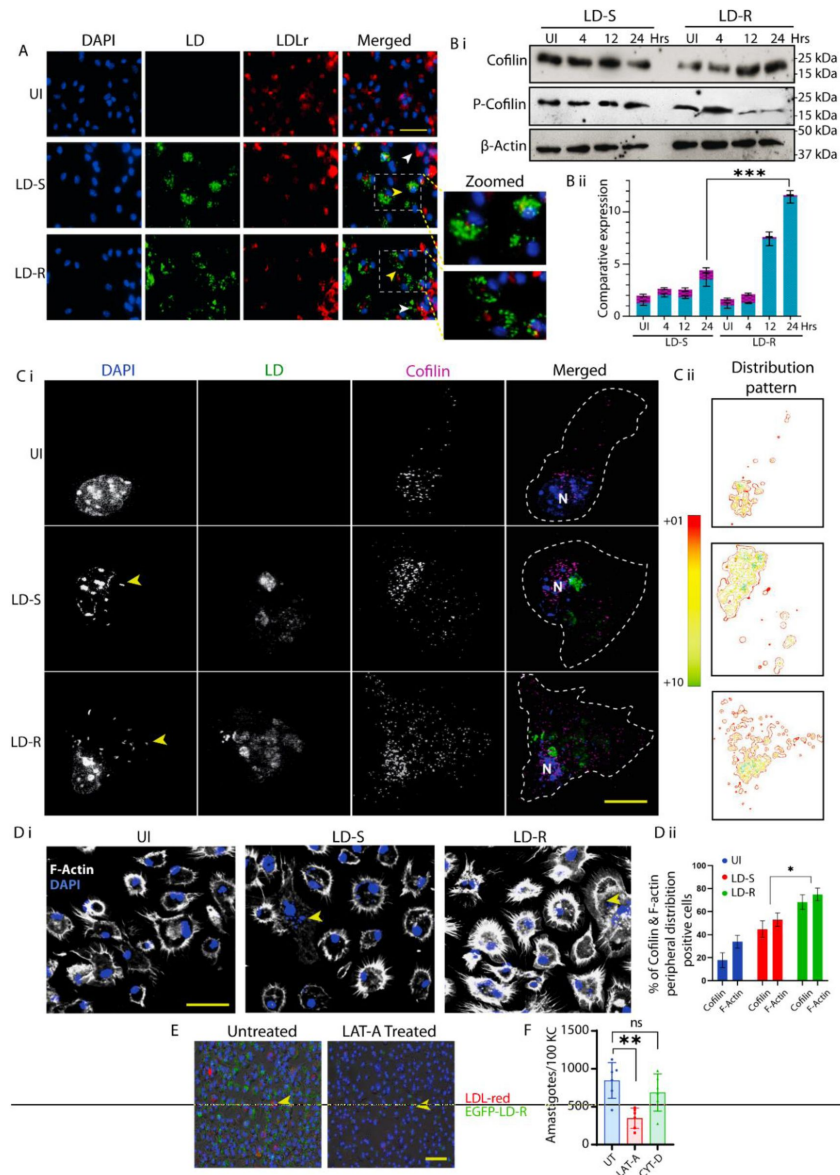
[41]. To confirm this possibility, we checked the status of Cofilin, an actin modulating protein, which serves as a marker for fluid phase endocytosis. Western blot analysis of infected-KCs revealed an increase in Cofilin levels with a concurrent decrease in its phosphorylated form as LD-R-infection progressed, indicating active fluid phase endocytosis (Figure 4Bi, ii). In contrast, LD-S-infection did not induce such changes (Figure 4B i, ii). Interestingly, this decrease in Cofilin phosphorylation was also observed in LD-R-infected-PEC (Figure S4C i, ii), signifying that enhanced fluid phase endocytosis may be a conserved mechanism for facilitating LDL-influx in LD-R-infected-MFs. This enhanced fluid phase endocytosis in response to LD-R-infection was further confirmed by structural illumination imaging (SIM), which showed increased Cofilin turnover with dispersed distribution in LD-R-infected-KCs, leading to enhanced F-Actin rearrangement and protrusion formation [42], which was absent for uninfected or LD-S-infected-KCs (Figure 4Ci, ii and Figure 4Di, ii). Finally, we infected the KCs with GFP expressing LD-R for 4Hrs, washed and allowed the infection to proceed in presence of fluorescent red-LDL and Latrunculin-A, a compound which specifically inhibits fluid phase endocytosis by inducing actin depolymerization [41]. Real-time fluorescence tracking demonstrated that Latrunculin-A treatment not only prevented the uptake of fluorescent red-LDL but also severely impacted intracellular proliferation of LD-R amastigotes (Movie 2A and 2B and Figure 4E). In contrast, treatment with Cytochalasin-D, which alters cellular F-actin organization but does not affect fluid phase endocytosis [41], had no effect on the intracellular proliferation of LD-R (Figure 4F and Figure S4D).

## 7. Endocytosed-LDL traffics and fuses with and LD-R-PV with increased proficiency

Typically, LDL-vesicles fuses with lysosomes upon entry into MFs, leading to subsequent transfer of LDL-derived cholesterol into the endoplasmic reticulum and cytoplasm [43]. As we have already confirmed an increased influx with higher presence of endocytosed-LDL vesicles around LD-R amastigotes along with significantly higher accumulation of cholesterol and fatty acids in LD-R-PV as compared to LD-S-PV (Figure 2), we hypothesized that the mode of trafficking of endocytosed- LDL might differ between LD-S and LD-R-infection. LD-PV are modified phagolysosomes [35, 44], and we observed an increased convergence of LDL-vesicles with LAMP-1 positive LD-R-PV even at 4Hrs p.i, while such convergence was not noticed in LD-S-infected-KC (Figure 5A i, ii). Structured Illumination Microscopy confirmed a complete convergence of LDL-vesicles with LD-R- PV at 24Hrs p.i, while LD-S-PV showed only partial convergence with LDL-vesicles even at this late infection point (Figure 5B i, ii). Interestingly, while, Western blot analysis revealed minimal expression of LAMP-1 in response to LD-S-infection at 4Hrs p.i, which increased considerably as the infection progressed (Figure 5C), however, a considerable amount of LAMP-1 expression was observed for LD-R-infected-KCs from 4Hrs p.i, which also increased as infection progresses (Figure 5C ii), indicating that a higher fusion of LDL-vesicles in LD-R-PV may be accompanied by increased lysosomal biogenesis required for degrading LDL into cholesterol. Importantly, Filipin staining of EGFP-LD-S and EGFP-LD-R-infected-KCs at 24Hrsp.i confirmed enriched cholesterol staining around LD-R amastigotes with significantly higher co-localization coefficient for LD-R amastigotes (Figure 5D i, ii, iii).

## 8. LD-R-infection suppress release of metabolized cholesterol from PV

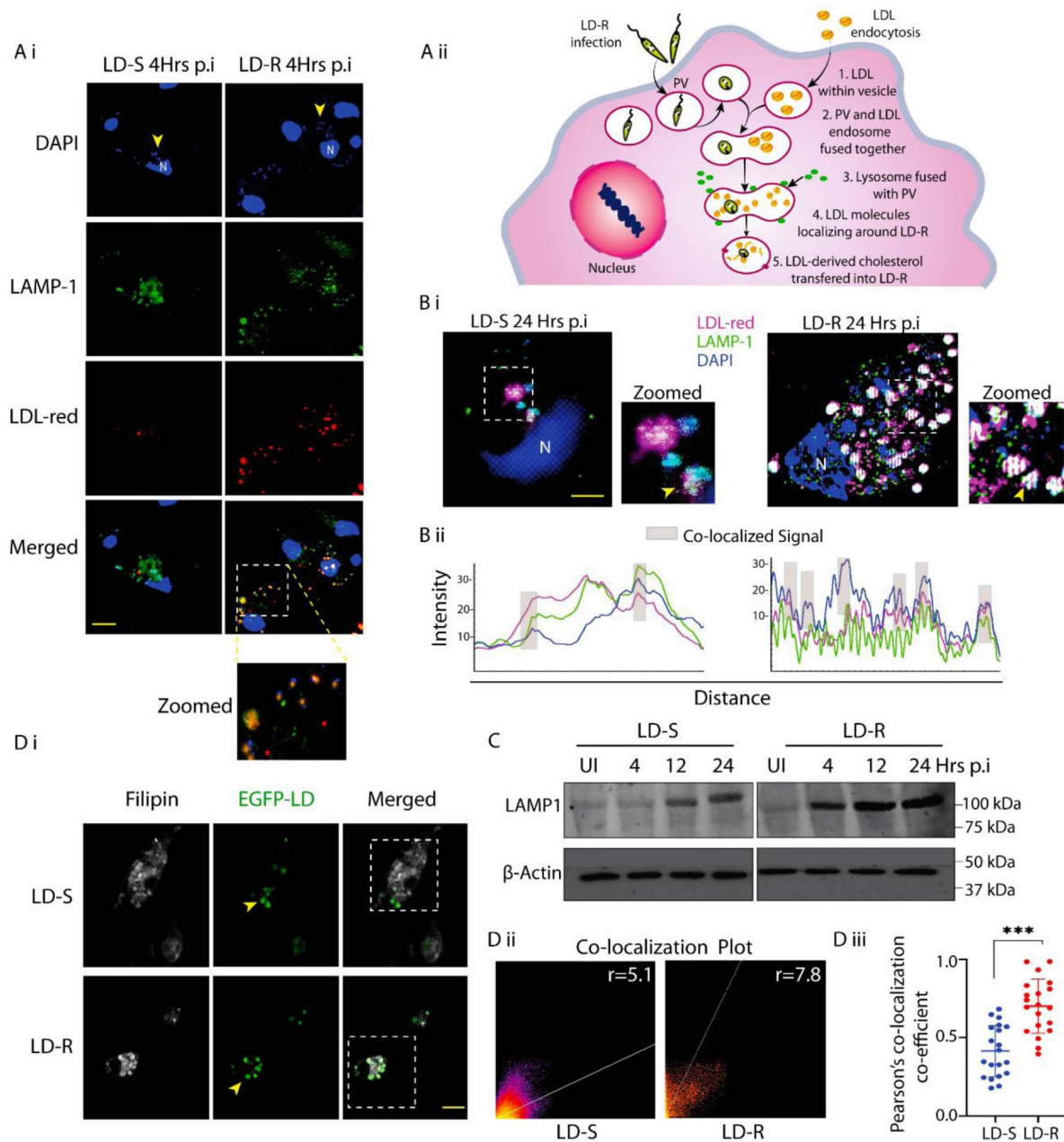
Once LDL is broken down to cholesterol by lysosomal acid hydrolases, metabolized cholesterol is released back into cytosol from LDL-vesicles by two transmembrane proteins namely NPC-1 and NPC-2 [43]. We suspected that LD-R must prevent or delay this process as this will limit cholesterol accessibility to replicating LD-R amastigotes. Analysis of available RNAseq (INRP000146) data for uninfected, LD-S and LD-R-infected-PEC (Figure 6A) revealed a suppression of *npc-1* expression along with its transcription factor *sreb2* [45], in response to



**Figure 4.**

### Increased LDL uptake in LD-R-Infected-MFs through receptor independent fluid phase endocytosis.

**(A)** Immunofluorescence images of LDLr expression in LD-infected-KCs. Scale bar 40µm. Enlarged view of A showing infected-KC with low LDLr expression (yellow arrow) compared to neighboring uninfected KC (white arrow) in the merged image. **B (i)** Cofilin and phosphorylated-Cofilin expression by western blot in LD-infected-KCs. **B (ii)** Graphical representation of comparative expression of Cofilin and phosphorylated-Cofilin with respect to  $\beta$ -actin for B (i). **C (i)** SIM images showing the Cofilin distribution in LD-infected-KCs. Yellow arrow represents intracellular LD, white dotted line in the merged image demarcates cell boundary. Scale bar 5µm. **C (ii)** Colour spectrum representing Cofilin distribution pattern of C (i) with green representing maximum and red representing minimum distribution. **D (i)** Confocal images representing Filamentous actin (F-actin) protrusions in LD-infected-KCs. Scale bar 20µm, with yellow arrow representing LD amastigotes. Individual experimental condition is represented in pseudo-colour, with uniform high contrast to clearly represent the F-actin protrusions. **D (ii)** Graph representing Cofilin and F-actin distribution in KCs infected with LD or left uninfected. **(E)** Still images from live cell video microscopy representing co-localization of LD-R with LDL in presence or absence of Latrunculin-A (LAT-A). Scale bar 60µm. Yellow arrow represents LD-infected-KCs with LDL (red). **(F)** Graphical representation (N=6) of amastigote count in 100 LD-R-infected-KCs treated with either LAT-A, or Cytochalasin (CYT-D) or left untreated. Data are presented as mean  $\pm$  SEM. \*\* signifies p value <0.001, \* signifies p value <0.05, n.s non-significant.



**Figure 5.**

### Endocytosed-LDL fuses with LD-R-PV to provide Cholesterol to LD-R amastigotes.

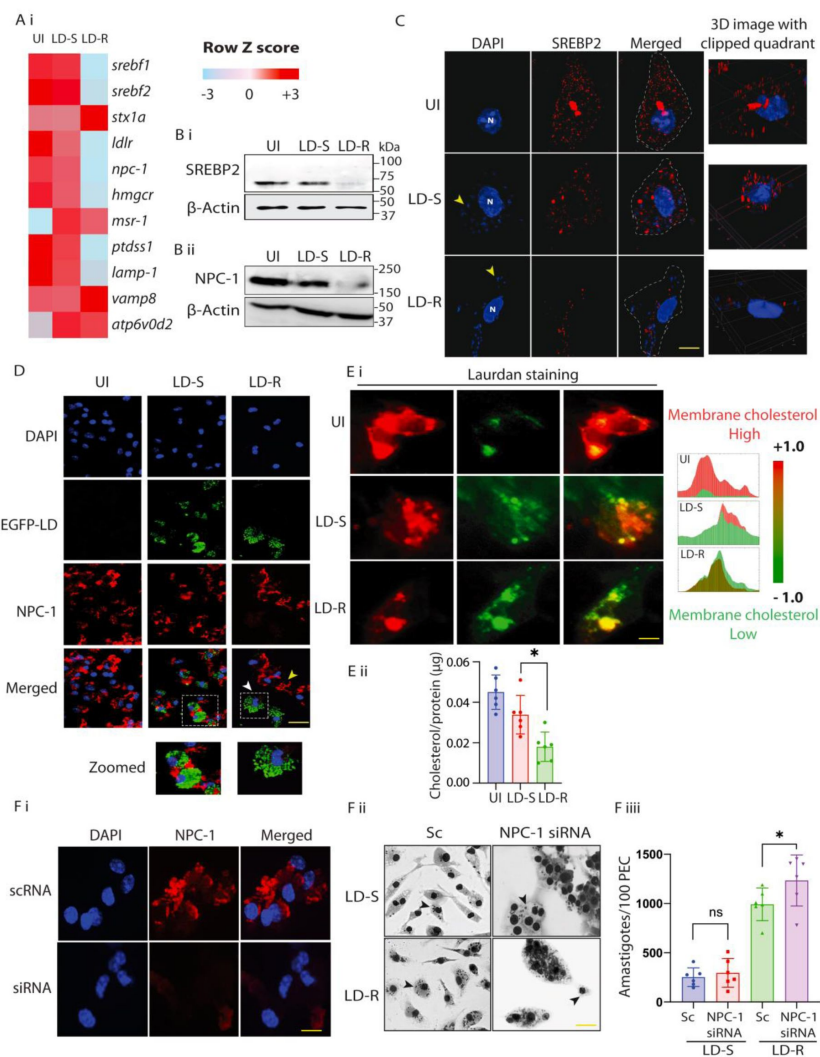
**(A)** Confocal images of representing LDL-vesicles, with LAMP1 positive lysosomal vesicles in LD- infected-KCs 4Hrs p.i. Yellow arrow represents LD nucleus, white dotted box marks the region further cropped and zoomed to show fusion of lysosomal vesicles with LDL and LD-R. Scale bar 10µm. **A (ii)** Schematic representation showing trafficking of LDL-vesicle towards the LD-R-PV. **B (i)** SIM images representing convergence of LDL-vesicles with LD amastigotes and LAMP1 in 24Hrs with LD- infected-KCs. White dotted box marks the region further cropped and Zoomed to clearly show the convergence (of LD amastigotes with LDL and LAMP-1 (white vesicles shown by yellow arrow), Scale bar 5µm. **B (ii)** Intensity plot representing of co-localization signals for B (i). **C** Western blot analysis showing LAMP1 expression in LD-infected-KCs from early to late hours. **D (i)** Filipin staining of EGFP-LD-infected-KCs. Filipin (white) and LD (green), yellow arrow represents intracellular LD. Scale bar 10µm. White dotted line marked cells used to generate co-localization plot. **D (ii)** Co-localization signals of LD (EGFP and Filipin represented by the dot plot. Pearson co- localization coefficient value ( $r$ ) represented for LD-S and LD-R-infected-KC. **D (iii)** Graph representing Pearson's coefficient for EGFP-LD-infected-KCs (N=19). Statistical significance in the observed differences was determined through ANOVA utilizing GraphPad Prism software (version 9.0). \*\*\* signifies  $p$  value < 0.0001.

LD-R-infection. Western Blot analysis of LD-S and LD-R-infected-KCs confirmed this low expression of SREBP2 and NPC-1 in LD-R- infected-KC 24Hrs p.i (**Figure 6Bi, ii**). More interestingly, Confocal imaging not only confirmed this low expression of SREBP2, but even in those cases of LD-R-infected-KCs where we did observe some residual expression of SREBP2, it was entirely restricted to cytoplasm, unlike LD-S or uninfected-KCs where SREBP2 did get translocated in the nucleus (**Figure 6C**). For NPC-1, immunofluorescence imaging comparing adjacent uninfected and infected-KCs harboring LD-S or LD-R, clearly confirmed unlike LD-S, LD-R specifically suppresses NPC-1 to prevent release of LDL-derived cholesterol from fused vesicles making it accessible to amastigotes (**Figure 6D**). We confirmed this observation by performing a Total Internal Reflection Microscopy (TIRF) which clearly showed a low plasma membrane cholesterol for LD-R-infected-KCs as compared to LD-S- infected-KCs (**Figure 6Ei**) and subsequently revalidated this by quantifying the cholesterol content in isolated membrane fraction for individual experimental set as mentioned above (**Figure 6E ii**). Finally, siRNA mediated knock down of NPC-1 in PEC (due to failure of successful KD generation in KCs), resulted in significant increase in amastigote load in response to LD-R-infection, while no such change in amastigote load was observed for LD-S-infected NPC-1-KD-PECs (**Figure 6F i, ii, iii**).

## 9. LD-R-Infection specifically excludes oxidized-LDL in infected murine liver to limit inflammation

The increased accumulation of LDL within MFs typically triggers an inflammatory response [46]. This possessed an inherent conceptual dilemma to our observation as increased inflammatory response should be detrimental for LD-R amastigotes [47, 48]. To check this, we measured inflammatory IFN- $\gamma$  by co-culturing splenic T-cells isolated from LD-S or LD-R-infected mice, with LD-S and LD-R-infected-KCs, respectively (**Figure 7 upper left panel**). Surprisingly, our ELISA results showed a low IFN- $\gamma$  levels in the culture supernatant of LD-R-infected-KCs even after 24Hrs of co-culture, while for LD-S-infection a significant increase was noted at this point (**Figure 7Ai**). Similarly, stimulation of LD-infected splenocytes obtained from LD-S and LD-R-infected *Apoe*<sup>-/-</sup> mice with specific soluble leishmania antigens (SLA) of LD-S or LD-R also confirmed this restricted IFN- $\gamma$  response to LD-R-infection (**Figure 7A ii**), thus suggesting that although LD-R-infection results in increased accumulation of lipids in host MFs, it elicits a suppressed inflammatory response as compared to LD-S. Notably, accumulation of oxidized-LDL (ox-LDL), and not its neutral form has been reported to cause inflammatory response [46, 49]. LD-S or LD-R-infected- KCs incubated with ox-LDL showed a significant purple positive staining of ox-LDL only for LD-S- infected-KCs (**Figure 7B i**), with LD-R-infected-KCs seemingly inhibiting uptake of ox-LDL. Infecting KCs with higher number of LD-S or lower number of LD-R did not alter their ox-LDL uptake capabilities, clearly suggesting LD-R-infection inherently suppress ox-LDL uptake, irrespective of the number of intracellular amastigotes (**Figure 7B. ii**). Although, receptor of ox-LDL, MSR-1 [26], have shown a slightly lower expression in LD-R-infected-PEC in the *RNAseq* analysis (**Figure 6A**), however owing to reported dominant expression of *msr-1* in the liver [49], we checked expression of MSR-1 in the LD-S and LD-R-infected-KCs. Confocal images and Western blot analysis confirmed a significant lower expression of MSR-1 in LD-R-infected-KCs as compared to LD-S-infected-KCs (**Figure 7C i, ii**), thus explaining reduced staining for ox-LDL in response to LD-R infection. Furthermore, increased expression of *atp6v0d2* transcripts, implicated in LD-PV acidification and lipid utilization [35], was also noted to be slightly higher for LD-R-infected-PECs in the *RNAseq* (**Figure 6A**). However, western blot analysis showed a slightly increased expression of ATP6V0D2 in LD-R-infected-KCs as compared to LD-S-infected-KCs (**Figure 7D**), suggesting that it might also play a role in maintaining LD-PV acidity, facilitating lipid utilization and limiting ox- LDL uptake, as previously reported for *L. mexicana* infection [35]. Finally, immunostaining of liver sections from 28 days LD-S and LD-R-infected-mice confirmed a significant lower expression of MSR-1 in LD-R-infected liver with higher accumulation of neutral lipid droplets as compared to LD- S-infected murine liver (**Figure 7E i, ii**), thus confirming LD-R specifically accumulates neutral lipids by restricting MSR-1 expression.



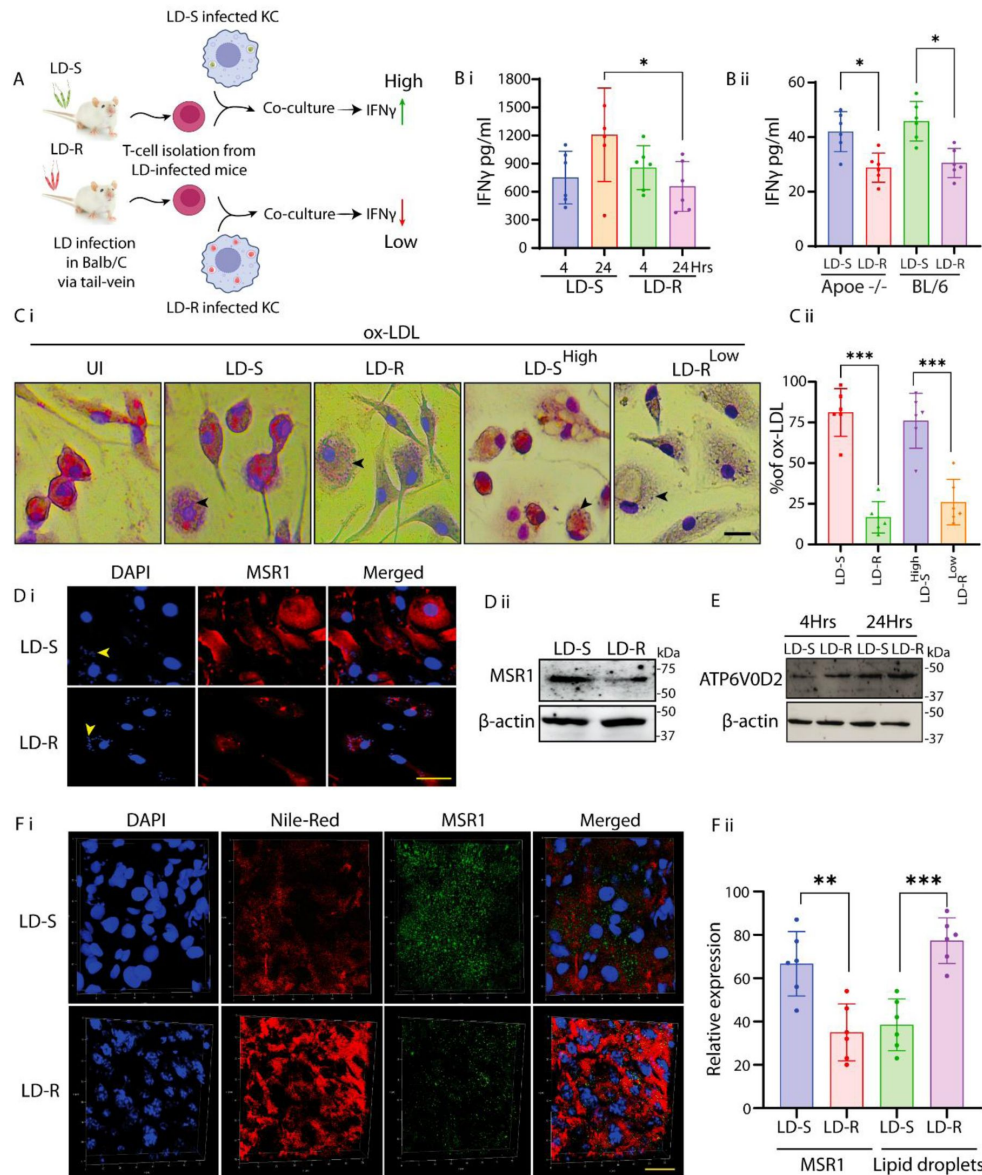


**Figure 6.**

### LD-R infection suppress NPC-1 to alter Cholesterol mobilization in infected MFs.

**(A)** 24Hrs LD-S and LD-R-infected-PECs were subjected to *RNAseq* analysis keeping uninfected-PECs as control. Heat map representing average expression of differentially expressed genes related to lipid metabolism between LD-S and LD-R infected-PEC. Scale represents median-centered counts (TPM). Row Z score data represented here. Expression of RNAs (right margins) is presented as centered and "scaled" [mean normalized log<sub>2</sub>(TPM + 1)]. **B (i)** Expression of SREBP2 and **B (ii)** NPC-1 in LD-infected and uninfected KCs by Western blot with  $\beta$ -actin as control. **(C)** Intracellular localization of SREBP2 in response to LD-infection in KC 24Hrs p.i. White dotted line represents cell periphery with yellow arrow representing intracellular LD and N representing host nucleus. 3D image represented in the right most panel with one quadrant clipped to confirm nuclear translocation of SREBP2. Scale bar 5  $\mu$ m. **(D)** Immunofluorescence showing NPC-1 expression in LD-infected-KCs. White arrow representing NPC-1 expression in LD-R infected-KCs and yellow arrow representing NPC-1 in adjacent uninfected-KCs in the merged panel. White dotted box marks regions further cropped and shown in the bottom panel with enlarged view (Zoomed). Scale bar 20  $\mu$ m. **E (i)** TIRF microscopic images of Laurdan stained LD-infected and uninfected KCs with two different spectra (488nm, 594nm), changes from red to green represents gel to fluid phase transition of the host plasma membrane. Colour distribution pattern corresponding to the TIRF images representing each experimental condition is presented in the right most panel. Scale bar 5  $\mu$ m. **F (i)** Expression of NPC-1 determined by immunofluorescence in PECs transfected with NPC-1 siRNA or scrambled (sc) siRNA. Scale bar 5  $\mu$ m. **F (ii)** Giemsa-stained images of LD-infected NPC-1 knockdown-PECs with scrambled siRNA control (Sc). Scale bar 20  $\mu$ m. Images are represented in grey scale with increased contrast to represent LD nucleus (black arrow). **F (iii)** Graphical representation of number of intracellular LD-amastigotes as in F (ii). Results show the counting of 100 infected-PEC from 6 independent replicate. \* signifies p value < 0.05, n.s non-significant.





**Figure 7.**

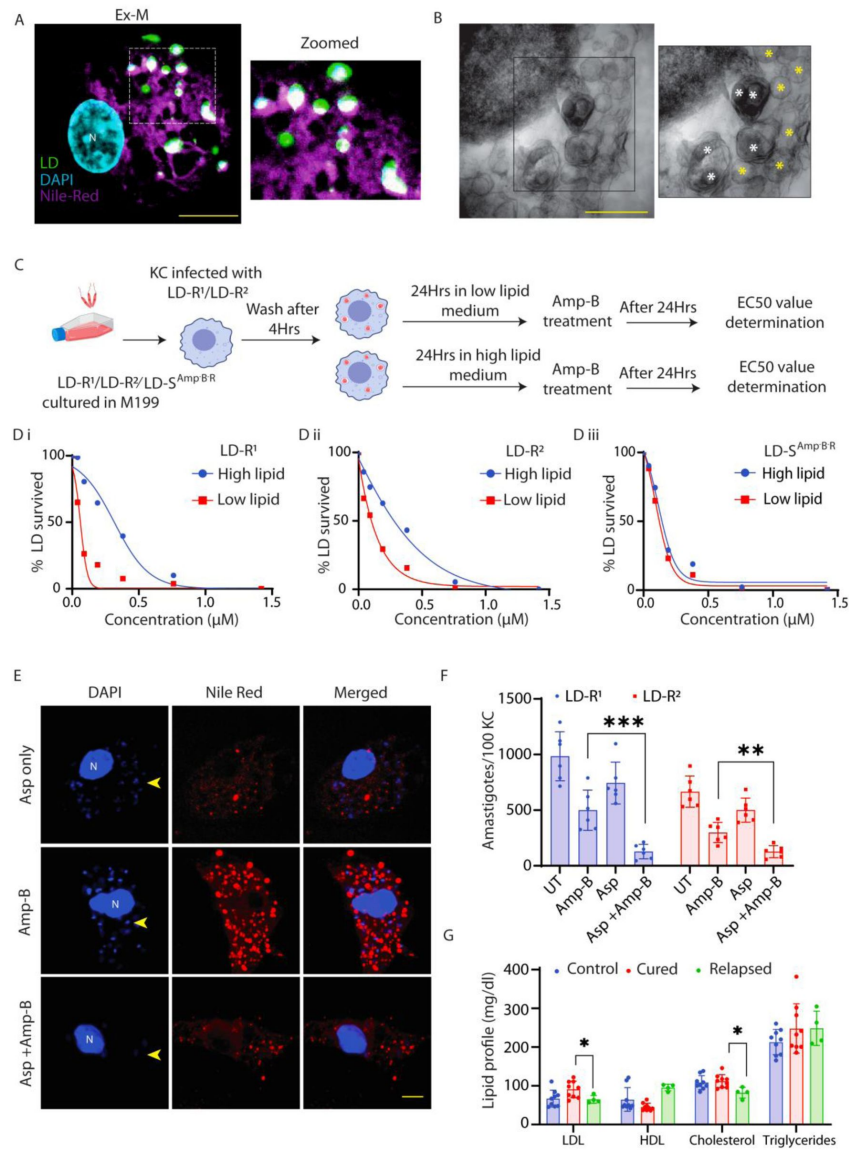
### LD-R-infected-MFs selectively excludes ox-LDL to suppress inflammatory response in host.

**(A)** Scheme representing ex vivo co-culture experiment performed by isolating T-cells from LD-infected mice with in vitro LD-infected-KCs **(B i)** IFN- $\gamma$  level in the supernatant of LD-infected-KCs co-cultured with the T-cells isolated from spleen of Balb/c mice infected with LD- S or LD-R 28 days p.i. Data represented as mean  $\pm$  SD from three independent experiments. **(B ii)** IFN- $\gamma$  level in the supernatant of splenocytes cultures from LD-infected-BL/6 and *Apoe*<sup>-/-</sup>-mice stimulated with SLA (Soluble *Leishmania* antigen) specific for LD-S or LD-R. Data are presented as mean  $\pm$  SD from six independent mice. **(C i)** ox-LDL incubated LD-infected-KCs was fixed and stained with Oil-red-O and Haematoxylin. LD-S<sup>High</sup> represents MOI (1:20) and LD-R<sup>Low</sup> represents MOI (1:5). Scale bar 10 $\mu$ m. Black arrow represents LD nucleus. **(C ii)** Graph representing % of ox- LDL positive MFs under different experimental conditions as represented in C (i). **(D i)** Confocal image showing MSR-1 in LD-infected-KCs. Yellow arrow represents LD-amastigotes while N represents host nucleus. Scale bar 20 $\mu$ m **(D ii)** Expression of MSR1 detected by Western blot of LD-infected-KC with  $\beta$ -actin as loading control. **(E i)** Expression of MSR1 and neutral lipid droplet accumulation assayed by Confocal microscopy from the cryo-sectioned liver of 28 days LD- infected-mice. Representative image showing Lipid droplet marked in red and MSR1 in green, nucleus in blue. Scale bar 40 $\mu$ m. **(E ii)** Graphical representation of % positive Lipid droplets accumulation and MSR-1 expression under different experimental condition as in E (i). Data collected from N=6 mice. \*\*\* signifies p value < 0.0001, \*\* signifies p value < 0.001, \* signifies p value < 0.05.

## 10. Lipid droplets accumulation decreases

### Amphotericin-B responsiveness in LD-R amastigotes

Increased LDL-uptake has been reported to cause high lipid droplets formation [50, 51], and lipid droplets formation has been reported to play critical role in intracellular proliferation of several pathogens, including LD-amastigotes [52–54]. As we have already observed significant accumulation of lipid droplets in LD-R-infected murine liver (Figure 7), we further investigated its role in response to LD-S and LD-R-infection. Lipid droplets staining of LD-S and LD-R-infected-KCs revealed significantly higher lipid droplets accumulation around LD-R amastigotes (Figure S5A and Movie 3). For better visualization, we expanded [55] EGFP-LD-R-infected-KCs and again performed lipid droplets staining on them which confirmed assimilation of neutral lipid into LD-R amastigotes (Figure 8A). This was further confirmed by performing TEM in LD-R-infected-KC which showed Lipid droplets accumulating around LD-R amastigotes (Figure 8B). Lipid accumulation has been previously reported to cause Amphotericin-B unresponsiveness among pathogens by hindering drug penetration [23]. Interestingly, we noted that two LD-R-strains (BHU575 and BHU814) which we have included in this study also shows reduced responsiveness against Amp-B, a second-line-antileishmanial [56], and we validate this by determining their EC<sub>50</sub> against Amp-B in LD-infected-KCs, which showed an EC<sub>50</sub> of 0.39  $\mu$ M and 0.31  $\mu$ M for LD-R<sup>1</sup> and LD-R<sup>2</sup> amastigotes respectively, as compared to EC<sub>50</sub> of 0.03  $\mu$ M for LD-S<sup>1</sup> and 0.032  $\mu$ M LD-S<sup>2</sup>. We suspected that significant lipid droplets accumulation around LD-R amastigotes might be responsible for their reduced susceptibility towards Amp-B by limiting drug accessibility. To test this possibility, we performed independent infection with two LD-R strains in low lipid and high lipid medium and access their EC<sub>50</sub> against Amp-B. Low lipid medium results in significant drop in lipid droplets accumulation within LD-R-infected-KCs (Figure S5B) and also resulted in enhanced susceptibility for both the LD-R strains against Amp-B with EC<sub>50</sub> reduced to 0.39  $\mu$ M to 0.16  $\mu$ M for LD-R<sup>1</sup> and 0.31  $\mu$ M to 0.06  $\mu$ M for LD-R<sup>2</sup> as compared to high lipid medium (Figure 8C i, ii). Interestingly, infection with lab-generated Amphotericin-B-unresponsive LD promastigotes (LD-S<sup>Amp-B-R</sup>) without primary unresponsiveness towards antimony did not result in any significant lipid droplets accumulation around LD-R amastigotes (Figure S5A) and neither showed any significant alteration in EC<sub>50</sub> between low (0.09  $\mu$ M) and high lipid medium (0.11  $\mu$ M) (Figure 8C iii). This observation clearly suggests that this high LDL-influx along with higher lipid droplet accumulation responsible for increased unresponsiveness of LD-R isolates towards Amp-B is a specific attribute of LD isolates with primary unresponsiveness against antimony and may vary significantly from lab generated Amp-B unresponsive strains. Previously, Aspirin (Asp), a lipid droplet blocker has been reported to block lipid droplet formation in *T. cruzi* and LD-infected-MFs [52, 57]. We also observed a drastic drop in lipid droplets formation in LD-R-infected-KCs treated with Aspirin as compared untreated LD-R-infected-KCs (Figure S5C). Interestingly, while treatment with Aspirin alone did not significantly reduce the amastigote load in LD-R-infected-KCs, combining Aspirin with previously determined low dose of Amp-B (0.1  $\mu$ M), completely restored Amp-B responsiveness against intracellular LD-R amastigotes (Figure 8E, F). This suggests that the enhanced uptake of LDL leading to lipid droplet accumulation might be a primarily responsible for increasing Amp-B unresponsiveness among clinical LD-R isolates. Importantly, this observation is consistent with findings from human VL patients from the endemic region of Bihar, India, where those who relapsed after Amp-B treatment had significantly lower serum LDL and cholesterol levels compared to Amp-B treatment responders (Figure 8G and Table 2).



**Figure 8.**

### Lipid droplets accumulation around LD amastigotes is inversely related with susceptibility towards Amphotericin-B.

**(A)** Ex-M image showing lipid droplet assimilation in LD-R amastigotes. White dotted region further cropped and shown in enlarged panel (Zoomed). Nile red (magenta), LD (green), nucleus (cyan). White spots represent Lipid droplets assimilated in LD-R and Host nucleus is represented as N. Scale bar 5μm. **(B)** TEM images showing lipid droplets around the LD-R amastigotes. Black box further magnified and represented in right panel. White \* representing LD-R amastigotes, yellow \* showing lipid droplets. Scale bar 2μm. **(C)** Schematic representation representing experimental scheme used to determine EC50 against Amphotericin-B under different experimental conditions. **D (i) and (ii)** Determination of EC50 value against Amphotericin-B in low lipid and high lipid media by performing infection with two independent LD-R strains. **D (iii)** Determination of EC50 value against Amphotericin-B in KC infected with LD-S<sup>Amp-B-R</sup>. Intracellular amastigotes were counted through Giemsa staining. **(E)** Confocal images showing lipid droplets accumulation around intracellular amastigotes with Aspirin (5 μM) or Amp-B (0.36 μM) or with a combination of Aspirin (5μM) and Amp-B (0.11μM) at 48Hrs p.i **(F)** Determination of amastigote load in KCs infected with two independent LD-R isolates in different experimental conditions either untreated or treated with Aspirin (5 μM) or Amp-B (0.36 μM) or with a combination of Aspirin (5μM) and Amp-B (0.11μM) at 48Hrs p.i **(G)** Comparative Lipid profile of VL patients (Cured and Relapsed) in response to Amp-B treatment along with healthy individuals from endemic region of Bihar, India. \*\*\* signifies p value < 0.0001, \*\* signifies p value < 0.001, \* signifies p value < 0.05.

## Discussion

Our findings revolve around understanding how antimony-resistant *Leishmania donovani* (LD-R) has evolved as a persistent strain by adopting to a strategy of better utilization of host derived lipids to sustain their own aggressive intracellular proliferation. Intracellular pathogens often alter the host's immune-metabolic state for better survival [58, 59]. Our observations pointed out that high organ parasite load linked with LD-R-infection does not rely only on initial infectivity as commonly believed to date [7], rather is indicative of a selective higher intracellular proliferation in host. We observed LD-R amastigotes scrutinize and access specific type of host lipoproteins to support their aggressive proliferation as seen in clinical infection (Figure 1, S1) [15]. Previously increased amastigote load in VL patients has been inversely linked with their serum cholesterol level [29], and we also observed that mice infected with LD-R isolates had significantly lower HDL, LDL, and cholesterol compared to LD-S-infected mice (Figure 1).

LD amastigotes are lipid auxotrophs [11, 14], and this severe dyslipidemia in response to LD-R-infection can be logically interpreted in terms of LD-R's increased need to support their aggravated intracellular proliferation in the infected-host. In liver-resident macrophages (Kupffer cells), primary sites for LD infection and host cholesterol synthesis, we observed significantly higher number of LD-R amastigotes and parasitophorous vacuoles formation (PV), compared to LD-S-infection, indicating LD-R's better capacity to utilize host cholesterol (Figure 1, S1). Also, in hyperlipidemic *Apoe*<sup>-/-</sup> mice, LD-R-infection led to organ parasite overload compared to LD-S-infection, confirming LD-R's increased dependence on extracellular lipid sources to sustain its aggressive proliferation (Figure 2). Thus, next we focused on investigating LD-R's ability to differentially regulate host's immune-metabolic status. Previous reports suggested that LD with altered drug-responsiveness can modulate their host differentially resulting in different disease outcome [3, 60]. Our results confirmed that while both LD-R and LD-S promastigotes initially quench cholesterol from the host plasma membrane similarly (Figure S2), LD-R amastigotes accumulate significantly more cholesterol and fatty acids in their PV by specifically promoting increased endocytosis of LDL as the infection progresses (Figure 3, S3). This possibly also explains the similar initial attachment and infectivity of LD-S and LD-R promastigotes with the host MFs again indicating towards their ability to infect their host equally (Figure 1, S1, Movie 1).

Interestingly, this LD-R-induced LDL-endocytosis is independent of the host's LDL receptor and relies on fluid-phase endocytosis, involving significant actin cytoskeleton reorientation in LD-R-infected-MFs (Figure 4, S4). It is worth mentioning that LD-R-infection has been reported to induce high levels of IL-10 [3], which is linked with enhanced fluid-phase endocytosis [41].

Importantly, we also noted an increased convergence of endocytosed LDL-vesicles with lysosomal associated marker protein (LAMP-1), along with higher LAMP-1 expression around LD-R amastigotes as compared to LD-S ones (Figure 5). Lysosomal acid hydrolases metabolize LDL into bioavailable cholesterol, and higher fusion of lysosomal vesicles with LD-R-PV explains the significant Filipin-cholesterol staining around LD-R amastigotes (Figure 5). These findings not only confirm that endocytosed-LDL can be better utilized by LD-R amastigotes but probably also indicates towards a rapid maturation of LD-R-PV due to increased LDL-vesicle fusion [13, 35].

Although available *RNAseq* analysis (Figure 6) did not support this increased expression of *lamp-1* in the transcript level, it did reflect a notable upregulation of vesicular fusion protein (VSP) *vamp8* and *stx1a* in response to LD-R-infection. LD infection can regulate LAMP-1 expression, and the role of VSPs in LDL-vesicle fusion with LD-R-PV is worthy of further investigation. Additionally, *RNAseq* analysis also showed suppressed expression of *npc-1* in LD-R-



infected-MFs. NPC-1 is crucial for mobilizing vesicular cholesterol to the cytoplasm and plasma membrane [43], and its suppressed expression correlates with higher cholesterol availability for aggressive LD-R proliferation (Figure 6).

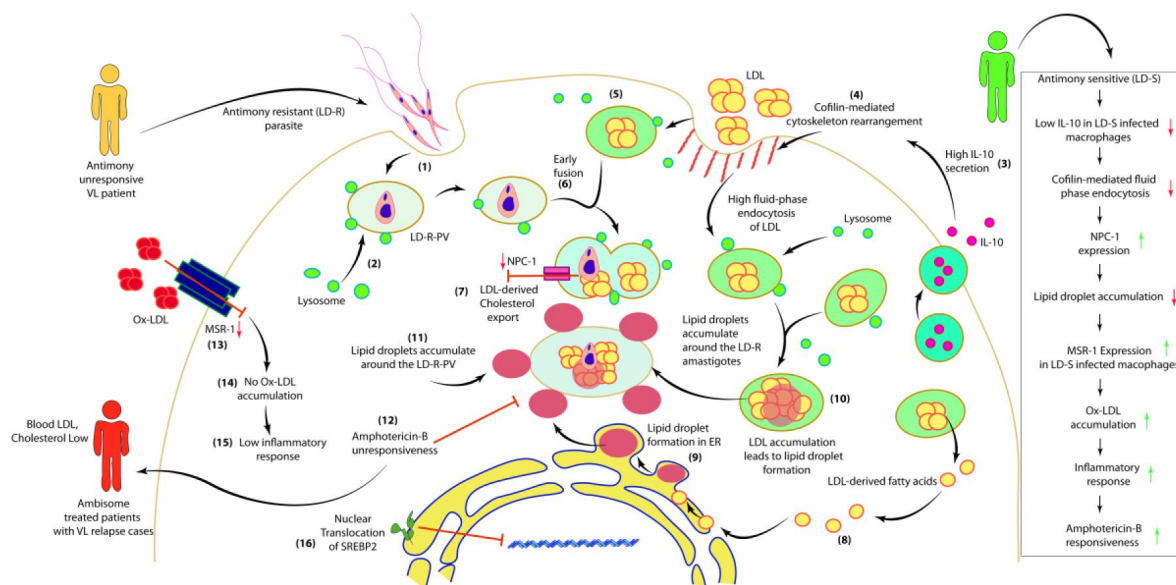
Besides LDL endocytosis, *de novo* cholesterol synthesis could also serve as a lipid source for intracellular LD amastigotes [61]. Surprisingly, LD-R-infected-MFs showed suppressed expression of SREBP2, with minimal nuclear translocation compared to LD-S-infection (Figure 6), consistent with reports of a shutdown in *de novo* cholesterol synthesis in MFs with high endocytosed LDL accumulation [62]. Notably, SREBP2 regulates NPC-1 expression [45] and has been and has been shown to get activated in response to LD-S infection [62].

Surprisingly, LD-R-Infected-MFs showed minimal inflammatory response despite high lipid accumulation, due to their ability to successfully exclude uptake of oxidized LDL (ox-LDL). Low expression of MSR-1, a scavenger receptor of ox-LDL, in the liver of LD-R-infected mice explained this observation, leading to the accumulation of neutral lipid droplets around LD-R amastigotes (Figure 7, S5, Movie 3). Expansion Microscopy (ExM) [55] along with TEM confirmed significant assimilation of these neutral lipids into LD-R amastigotes (Figure 8). Several intracellular pathogens like *Mycobacterium*, *Toxoplasma*, HCV have been reported to transfer stored triacylglycerols and sterol esters from lipid droplets to support their proliferation [10, 63, 64]. While the understanding of lipid uptake and metabolizing genes in LD is limited, a recent genome-wide CRISPR screen in *L. Mexicana* [65], have identified such potential candidates “essential” for parasite survival. Close proximity lipid droplets with LD-R amastigotes makes it tempting to speculate existence of lipid droplets contact sites [13], or possible copy number variations in these lipid metabolizing genes [66], providing survival advantages to LD-R. This speculation is further supported by the fact that although clinical LD-R isolates with neutral lipid droplet accumulation in infected MFs displayed increased unresponsiveness to Amphotericin-B (Amp-B), lab-generated Amp-B unresponsive promastigotes failed to do so in infected MFs (Figure 8). Fact that LD-R amastigotes regained their responsiveness towards Amp-B in low lipid medium or in presence of Aspirin [52], further confirms that eliminating lipid droplets around LD-R amastigotes might be a weapon of choice to combat treatment failure against Amp-B [27, 28, 67]. Our results thus suggest that natural clinical LD-R-isolates have developed an inherent adaptability to gradually develop cross-resistance towards Amp-B, probably as an additional consequence of their ability to accumulate increased host derived lipids as a rapidly proliferating intracellular amastigote (represented graphically in Figure 9). Our work explains increasing Amp-B treatment failure among VL patients [27, 28, 67] and correlates with low serum LDL and cholesterol levels in patients relapsing after Amp-B treatment (Figure 9, Table 2).

## Methods

### Reagents

Laurdan (Merck), photoRED-LDL (Thermo Fisher Scientific), NBD Cholesterol (Sigma), Nile red (Sigma-Aldrich), Oil red O (Thermo Fisher Scientific), Filipin (Sigma-Aldrich), Collagenase (Gibco) were purchased. Tissue culture plasticware were purchased from NUNC (Roskilde, Denmark). Invitrogen Amplex-Red assay kit was purchased from Thermo Fisher Scientific. Primary antibodies required for the experiments were anti-ATP6V0D2 (ABclonal), anti-MSR1 (ABclonal), anti-LAMP1 (Thermo Fisher Scientific), anti-SREBP2 (Thermo Fisher Scientific), anti-NPC1 (Santa Cruz Biotechnology), anti-LDLr (Santa Cruz Biotechnology), goat anti-mouse IgG Alexa Fluor 488 (Thermo Fisher Scientific), goat anti-mouse IgG Alexa Fluor 594 (Thermo Fisher Scientific), goat anti-Rabbit IgG-HRP (Thermo Fisher Scientific), goat anti-Mouse IgG-HRP (Thermo Fisher Scientific),  $\beta$  actin (Cell Signalling Technology), CD11b (Cell Signalling Technology), Cofilin (Cell



**Figure 9**

### Scrutinized uptake of host lipids results in breakdown of Amphotericin-B responsiveness in clinical LD isolates with primary resistance to antimony.

(1-2) LD-R residing in membrane bound parasitophorous vacuoles (PVs), and (3-5) induce high fluid phase endocytosis of LDL by modulating actin depolymerization of host membrane through cofilin. (6-7) LDL containing vesicles fused with LD-R-PV which provides increased lipid access to sustain their aggressive intracellular proliferation. Vesicular cholesterol export protein, NPC-1 is downregulated in the fused membrane, preventing release of LDL derived cholesterol back to host cytoplasm. (8- 12) LDL derived cholesterol esters also recycled in the ER and form lipid droplets which ultimately accumulate in and around the LD-R-PV ultimately leading to the Amphotericin-B unresponsiveness. (13-15) LD-R control inflammatory response related to accumulation of oxidized lipids (Ox-LDL) by downregulating MSR-1 (16) High LDL influx also results in shutdown of *de-novo* cholesterol synthesis in LD-R infected host through restriction of nuclear translocation of SREBP2. Right panel, representing LD-S infected host with low LDL influx with high NPC-1 expression leading to compromised lipid droplet accumulation around its PV, thereby exhibiting high responsiveness against Amphotericin-B.

Signalling Technology), and phospho-Cofilin (Cell Signalling Technology). RhodaminePhalloidin reagent was purchased from Thermo Fisher Scientific and ELISA Kits for IFN- $\gamma$  were purchased from R&D Systems, Aspirin generic composition, SAG (Albert David)

## Animals

BALB/c, C57BL/6 and *Apoe*<sup>-/-</sup> mice were maintained and bred under pathogen-free conditions with food supplements and filtered water. All experiments were conducted in accordance to Institutional Animal Ethics Committee (IAEC), IIT Kharagpur guidelines India (IAEC/BI/1 55/2021) and approved by National Regulatory Guidelines issued by Committee for the Purpose of Supervision of Experiments on Animals, Ministry of Environment and Forest, Government of India.

## Ethical statement

In this study, all procedures involving human participants were conducted in accordance with the ethical standards of the 1964 Helsinki Declaration and approval of the “Institutional Human Ethical Committee” of ICMR-Rajendra Memorial Research Institute of Medical Sciences, Patna, India (Approval No: RMRI/EC/20/2020). Following information about the potential risks, benefits, and the investigational nature of the study consent from all study participants was obtained in the informed consent form.

## Parasite Cultures and Maintenance

*Leishmania donovani* (LD) isolates used in this study are natural clinical isolates obtained from the patients MHOM/IN/09/BHU569/0, MHOM/IN/2009/BHU575/0, MHOM/IN/10/BHU814/1, MHOM/IN/83/AG83, MHOM/IN/09/BHU777/0, MHOM/IN/80/DD8 [2, 3], and were maintained in BALB/c mice as per institute biosafety guidelines (BT/BS/17/27/97-PID). Susceptibility of each of these LD isolates against antimony were checked and confirmed against previously reported EC50 value for antimony before they were included in the study (This can be reported in a common table, **Table 1**). Lab generated Amp-B resistant strain (DD8-AmpB) strain was kind from Dr. Arun Kumar Haldar (CSIR-CDRI). Amastigotes were isolated by macerating the spleen of 21 days infected mice under sterile conditions, and subsequently transformed into promastigotes by maintaining in M199 (Gibco) supplemented with 5% (v/v) heat-inactivated fetal bovine serum (FBS) (Gibco) and 1% (v/v) Penicillin Streptomycin solution (Gibco) under drug selection.

## Cell Culture and Infection

Peritoneal exudate cells (PECs) were harvested from BALB/c mice as mentioned in already published protocol, with slight modifications [68]. After 48Hrs of 4% starch intraperitoneal injection, PECs were isolated and plated on 35-mm tissue culture petri dishes, sterile 22-mm square coverslips in 35-mm petri dishes or 24-well plates at a density of  $1 \times 10^6$ ,  $1.5 \times 10^5$  or  $1.5 \times 10^5$ , respectively, all in RPMI 1640 medium (Gibco) supplemented with 10% (v/v) heat-inactivated FBS, 1% (v/v) Penicillin Streptomycin solution. The cells were allowed to adhere for 48Hrs at 37°C in 5% CO<sub>2</sub> before the infection process.

## In vivo infection

For *in vivo* infections, healthy female C57BL/6 mice or *Apoe*<sup>-/-</sup> mice (4-6 weeks old and 20-25 g each) were injected via tail vein route with  $1 \times 10^7$  GFP-expressing *L. donovani* stationary phase metacyclic promastigotes (suspended in 100  $\mu$ l 1X PBS) as reported previously [69]. After infection mice were kept in atherogenic diet till sacrifice. Each group contained five mice that were sacrificed 11 days' post-infection (p.i). Splenocytes were isolated and GFP positive population was measured through flow cytometry. Each experiment was performed in triplicate.

Table 1

## EC50 against SAG (Antimonial) for the LD strains used in this study

| Strain Name           | EC50 value for SAG( $\mu$ g/ml) in PEC | EC50 value for SAG( $\mu$ g/ml) in KC |
|-----------------------|----------------------------------------|---------------------------------------|
| MHOM/IN/2009/BHU575/0 | 22.73 $\pm$ 3.7                        | 23.89 $\pm$ 2.1                       |
| MHOM/IN/10/BHU814/1   | 30.3 $\pm$ 7.2                         | 28.5 $\pm$ 5.3                        |
| MHOM/IN/83/AG83       | 1.52 $\pm$ 1.2                         | 3.3 $\pm$ 2.2                         |
| MHOM/IN/80/DD8        | 2.1 $\pm$ 1.2                          | 3.2 $\pm$ 1.9                         |
| DD8-AmpB              | 1.8 $\pm$ 2.1                          | 2.2 $\pm$ 1.3                         |
| BHU575-EGFP           | 21.45 $\pm$ 3.2                        | 22.81 $\pm$ 3.1                       |
| AG83-EGFP             | 2.1 $\pm$ 1.4                          | 2.4 $\pm$ 3.1                         |

Table 2

## Lipid profile of VL patients

| Serum Code | Age | Sex | Time of Blood Collect | Infection status               | Treatment received   | Patient status | LDL (mg/dl) | HDL (mg/dl) | Cholesterol (mg) | Triglycerides (mg/dl) |
|------------|-----|-----|-----------------------|--------------------------------|----------------------|----------------|-------------|-------------|------------------|-----------------------|
| RMRI N1    | 45  | F   | Before                | Control population from endemi | NA                   | NA             | 45.3501585  | 46.6365931  | 101.6949153      | 177.258567            |
| RMRI N6    | 70  | F   | Before                | Control population from endemi | NA                   | NA             | 110.855478  | 50.3762712  | 111.8644068      | 206.5420561           |
| RMRI N7    | 45  | F   | Before                | Control population from endemi | NA                   | NA             | 75.5620787  | 41.8711864  | 84.40677966      | 165.4205607           |
| RMRI N8    | 19  | M   | Before                | Control population from endemi | NA                   | NA             | 83.946038   | 62.9533123  | 100.3389831      | 224.9221184           |
| RMRI N11   | 37  | M   | Before                | Control population from endemi | NA                   | NA             | 72.4549921  | 120.914196  | 134.2372881      | 244.5482866           |
| RMRI N12   | 60  | M   | Before                | Control population from endemi | NA                   | NA             | 52.6307448  | 113.851735  | 92.54237288      | 180.6853583           |
| RMRI N13   | 30  | M   | Before                | Control population from endemi | NA                   | NA             | 66.7533281  | 48.3413249  | 138.3050847      | 236.4485981           |
| RMRI N14   | 22  | M   | Before                | Control population from endemi | NA                   | NA             | 47.8939778  | 49.6807571  | 100              | 260.7476636           |
| RMRI N15   | 37  | F   | Before                | Control population from endemi | NA                   | NA             | 46.4904913  | 45.9059937  | 108.8135593      | 219.6261682           |
| RMRI C1    | 20  | F   | Before                | Confirmed VL, Hospitalization  | Ambisome Single dose | Cured          | 110.389045  | 44.2700565  | 112.5423729      | 200.623053            |
| RMRI C2    | 62  | F   | Before                | Confirmed VL, Hospitalization  | Ambisome Single dose | Cured          | 121.738904  | 64.5514124  | 145.4237288      | 301.8691589           |
| RMRI C3    | 15  | M   | Before                | Confirmed VL, Hospitalization  | Ambisome Single dose | Cured          | 76.1839888  | 36.7463277  | 95.59322034      | 185.6697819           |
| RMRI C4    | 31  | M   | Before                | Confirmed VL, Hospitalization  | Ambisome Single dose | Cured          | 72.141573   | 41.1079096  | 116.9491525      | 241.7445483           |
| RMRI C5    | 52  | M   | Before                | Confirmed VL, Hospitalization  | Ambisome Single dose | Cured          | 98.1063202  | 48.1954802  | 117.2881356      | 282.2429907           |
| RMRI C6    | 18  | F   | Before                | Confirmed VL, Hospitalization  | Ambisome Single dose | Cured          | 68.8765449  | 36.7463277  | 96.61016949      | 200.623053            |
| RMRI C7    | 45  | F   | Before                | Confirmed VL, Hospitalization  | Ambisome Single dose | Cured          | 110.544522  | 55.6101695  | 124.4067797      | 381.9314642           |
| RMRI C8    | 32  | F   | Before                | Confirmed VL, Hospitalization  | Ambisome Single dose | Cured          | 94.8412921  | 43.0706215  | 106.1016949      | 232.3987539           |
| RMRI C18   | 66  | M   | Before                | Confirmed VL, Hospitalization  | Ambisome Single dose | Cured          | 61.4136236  | 35.9830508  | 91.86440678      | 203.4267913           |
| RMRI R14   | 8   | M   | Before                | Confirmed VL, Hospitalization  | Ambisome Single dose | Relapse        | 79.2935393  | 83.6333333  | 96.94915254      | 262.9283489           |
| RMRI R16   | 49  | M   | Before                | Confirmed VL, Hospitalization  | Ambisome Single dose | Relapse        | 56.9047753  | 102.279096  | 63.05084746      | 209.0342679           |
| RMRI R17   | 18  | M   | Before                | Confirmed VL, Hospitalization  | Ambisome Single dose | Relapse        | 59.7033708  | 101.079661  | 83.38983051      | 217.1339564           |
| RMRI R18   | 59  | M   | Before                | Confirmed VL, Hospitalization  | Ambisome Single dose | Relapse        | 63.4348315  | 94.319209   | 84.40677966      | 304.6728972           |



## Isolation of Kupffer cells

Kupffer cells (KCs) were harvested from mice livers following a previously published protocol, with slight modifications [70]. Wild-type BALB/c mice were anesthetized, and warm PBS containing 0.05% collagenase IV was used for liver perfusion. The livers were swiftly excised and minced into small fragments on an ice bath. Tissue digestion was performed using a 0.1% Type IV Collagenase solution in RPMI 1640 for 30 minutes at 37°C with intermittent shaking for every 10 minutes. Following digestion, KCs were separated from hepatocytes and other cellular components through sequential centrifugation steps and plated on well plates, as required, for 4Hrs in RPMI 1640 media supplemented with 10% (v/v) Penicillin Streptomycin and 10% (v/v) FBS. After 24Hrs of resting at 37°C in 5% CO<sub>2</sub>, the infection studies commenced. To assess the purity of the isolated KCs by flow cytometry, KCs were plated on a 6 well plate. After 4Hrs, KCs were scraped out, centrifuged at 300g for 5 minutes and stained with monoclonal anti-F4/80 PE conjugated antibody (BD Biosciences). The KCs were then washed and fixed with 2% paraformaldehyde for 5 minutes, analyzed using a flow cytometer (ARIA FACS DIVA), and further processed using FLOW JO software for data analysis. Purified KCs were infected with stationary phase sorted metacyclic LD promastigotes as mentioned previously. PECs and KCs are collectively designated as MFs (macrophages).

## Sorting of Metacyclic LD and infection in MFs

Metacyclic parasites were sorted through Beckman Coulter CytoFLEX cell sorter, as per manufacturer's protocol. The LD promastigotes were washed and resuspended in PBS before being subjected to light scatter analysis using flow cytometry. Sorted population were checked with forward scatter (FSC) versus side scatter (SSC) and dot plot was generated to represent the acquisition of 10,000 events using a FACS Aria II system. Required number of sorted metacyclic LD were used to perform subsequent MF and mice infections. MFs were infected with stationary-phase Flow sorted metacyclic LD promastigotes at a MOI 1:10 ratio (in some cases 1:5 and 1:20) for 4Hrs which was considered as the 0<sup>th</sup> point of the infection. MFs was then washed to remove any extracellular loosely attached parasites and incubated further as per experimental requirement.

## Intracellular amastigote load determination

To enumerate intracellular amastigote load, resting stage PECs and KCs on glass coverslips were infected with sorted metacyclic LD promastigotes of various strains at a ratio of 1:10, as described in previous protocols [3, 71]. Following a 4Hrs incubation, the PECs or KCs were washed and subsequently cultured for specific durations. Coverslips were retrieved after specific time points, rinsed with PBS, fixed using methanol, and stained with Giemsa. These prepared coverslips were then affixed onto glass slides and examined utilizing a light microscope (Olympus). Parasite quantification was conducted by counting amastigotes per 100 macrophages. Image processing related to **Figure 1A** and Figure S1 D was performed using Fiji by separating the spectral overlap between the light and dark stained region to Giemsa stained MFs. Next, for better visualization, pseudo colours were assigned to represent nucleus (white) and cytoplasm (red) with a black background. Colour separated images were then merged showing LD within PV.

## Raman Spectroscopy of LD-infected-KCs

KCs were plated on Raman-grade stainless steel plate and kept for 24Hrs in a humidified environment at 37°C with 5% CO<sub>2</sub>, followed by infection with LD as already described. At 24Hrs p.i., the KCs were washed using PBS, fixed with 2% paraformaldehyde and proceeded to perform Raman spectroscopy (Witec, alpha300R). The Raman grade stainless steel plates were positioned on the microscope stage and focused with a 100x air objective and measurements were performed using a 532 nm laser source producing a laser spot of approximately 0.3 µm with 20mW laser power. Spectral measurements were obtained through the Project FIVE interface (Witec, Germany) after setting the grating to 600 gratings/mm. The spectral resolution was set to 2.87 cm<sup>-1</sup>. The

Raman spectra were taken with integration time set to 5 seconds, and ten accumulations were selected after confirming its effectiveness on the KCs being studied. Baseline correction and other spectral denoising procedures were carried out using MATLAB 2017b (MathWorks), and the processed spectra were analysed as described previously [36, 38].

## Quantification of membrane cholesterol with Amplex Red assay kit

The KC membrane preparation was conducted following the method described previously [20], with slight modifications. KCs were ruptured by repeated freezing and thawing cycles followed by probe sonicating for 4 cycles with 30secs burst and 30sec gap. The resulting homogenate was then centrifuged at 900g for 10mins at 4°C. The supernatant was filtered through a nylon mesh (100 µm), while the pellet was discarded. The filtered supernatant was centrifuged at 100,000g for 1hr at 4°C, and resulting pellet, representing native membrane, was suspended in a buffer containing 50 mM Tris (Ph 7.4), 1 mM EDTA, 0.24 mM PMSF, and 10 mM iodoacetamide. The protein content of the membrane was quantified using Bradford reagent (Biorad), following manufacturer's protocol. The total cholesterol content was determined using an Amplex Red reagent kit [72], with the results expressed in moles of cholesterol per gram of protein.

## Parasitophorous vacuole (PV) Isolation and cholesterol measurement

24Hrs LD infected KCs were lysed by passing through a 22-gauge syringe needle to release cellular contents. Parasitophorous vacuoles (PV) were then isolated using a previously outlined protocol [73]. Cholesterol and protein contents with PV were determined by an Amplex Red assay kit and Bradford assay, respectively. Resulting data were represented as micrograms of cholesterol per microgram of protein.

## GC-MS analysis of LD-S and LD-R-PV

Following a 24Hrs infection period, KCs were harvested, washed with phosphate-buffered saline (PBS), and pelleted. Subsequent to this, PV isolation was carried out using the previously described method [74]. The resulting PV pellet was suspended in 20 ml of dichloromethane:methanol (2:1, vol/vol) and incubated at 4°C for 24hours. After centrifugation (11,000 g, 1 hour, 4°C), the supernatant was checked through thin layer chromatography (TLC) and subsequently evaporated under vacuum. The residue and pellet were saponified with 30% potassium hydroxide (KOH) in methanol at 80°C for 2 hours. Sterols were extracted with n-hexane, evaporated, and dissolved in dichloromethane. A portion of the clear yellow sterol solution was treated with N, O-bis(trimethylsilyl)trifluoroacetamide (BSTFA) and heated at 80°C for 1 hour to form trimethylsilyl (TMS) ethers. Gas chromatography/mass spectrometry (GC/MS) analysis was performed using a Varian model 3400 chromatograph equipped with DB5 columns (methyl-phenylsiloxane ratio, 95/5; dimensions, 30 m by 0.25 mm). Helium was used as the gas carrier (1 ml/min). The column temperature was maintained at 270°C, with the injector and detector set at 300°C. A linear gradient from 150 to 180°C at 10°C/min was used for methyl esters, with MS conditions set at 280°C, 70 eV, and 2.2 kV.

## Cytokine measurement

Supernatants from KCs infected with stationary-phase LD co-culture with T cells, isolated from the mice infected with LD. For detection of IFN- $\gamma$  from Apoe<sup>-/-</sup>, splenocytes isolated from LD infected mice and cultured with SLA. IFN- $\gamma$  levels in the supernatants were measured using a sandwich ELISA Kit (R&D Systems) as per manufacturer's protocol. Detection limit for these kits were 8pg/ml for IFN- $\gamma$ . Cytokine levels were determined by measuring the OD at 450 nm using a MultiSkan FC microplate photometer (Thermo Fisher Scientific). Data was demonstrated as mean  $\pm$  SD of all five individual experiments.

## Western Blot Analysis

Lysates of LD-infected PECs and KCs were prepared, and western blotting was performed for different proteins with endogenous control  $\beta$ -actin (Cell Signaling Technology). Blots were probed with specific antibodies. Binding of secondary HRP-labeled goat anti-rabbit or goat anti-mouse Abs was analyzed using SuperSignalR West Pico or West Dura Chemiluminescent substrate (Pierce).

## Confocal and Structural Illumination Microscopy (SIM)

PECs and KCs were cultured on 13-mm diameter glass coverslips at a density of  $2 \times 10^5$ , and infected with LD as mentioned previously. After varying time intervals, infected PECs or KCs were rinsed with PBS, fixed with 2% paraformaldehyde, and permeabilized using 0.2% Triton/PBS for 20mins on an orbital shaker. Non-specific binding was blocked with 2% BSA in 0.2% Triton/PBS for another 20mins on an orbital shaker. The PECs and KCs were then incubated with primary antibodies for 1Hr, followed by washing and staining with IgG Alexa Fluor 594 or IgG Alexa Fluor 488, as required. Mounting was performed using Fluoromount-G-DAPI (Thermo Scientific). Images were captured using Olympus confocal microscopy (FV3000) and super-resolution microscopy ZEISS (SIM) with a 63x objective. 3D image visualization processed through Fiji 3D viewer and 3D project plugin. Live cell microscopy was performed using ZEISS phase contrast microscope and ZEISS Apotome. MFs were plated on confocal dish and incubated with photoRED-LDL for live cell imaging of LDL uptake ([Movie2](#) and [Movie 3](#) [🔗](#)).

## Lipid droplet staining

LD-infected-KCs were first fixed using 2% paraformaldehyde (PFA) and subsequently permeabilized with 0.25% Triton X-100. Coverslips were then treated with Nile red (Sigma, diluted 1:100) in PBS for at least 1Hr to stain lipids, followed by DNA staining with DAPI. The coverslips were mounted onto glass slides using Fluoromount-G and visualized with confocal microscope (LEICA STELLARIS 5). 3D stacks were acquired to visualize distribution of lipid droplets. Images were processed and analysed, and stacks were merged to reconstruct 3D images using Fiji.

## Ultrastructure expansion microscopy

Ultrastructure Expansion Microscopy (U-ExM) was conducted on LD-infected-MΦs following established experimental protocols [\[55\]](#). Protein crosslinking was initiated by immersing the coverslip in a solution containing 1.4% formaldehyde, 2% acrylamide, and PBS for 5Hrs at 37°C. Monomer solution (19% sodium acrylate, 10% acrylamide, and 0.1% N,N'-methylenebisacrylamide in PBS) was mixed with 10% TEMED and 10% APS solution on ice and added to 6-well pates. Gelation was performed on ice, with cell-plated coverslip placed face down on the gelation solution in a humid chamber for 5 minutes at 37°C for 1 hour. The gel and coverslip were together transferred to a 6-well plate filled with denaturation buffer (200 mM SDS, 200 mM NaCl, and 50 mM Tris pH 9.0) and incubated face up for 15mins at room temperature under agitation, followed by incubation at 95°C for 1.5Hrs. The first expansion phase involved incubating the gel three times in ddH<sub>2</sub>O for 30mins each. Previously mentioned IFA steps were adopted for antibody staining. Images were captured using an Olympus confocal microscope (FV3000) equipped with a 63X 1.4 NA oil objective and Small Volume Computational Clearing mode was used to obtain deconvolved images. 3D stacks were acquired and analyzed using Fiji software.

## Transmission electron microscopy (TEM)

For transmission electron microscopy, PECs infected with the LD-R strain were fixed after 24 hours of infection using 2.5% glutaraldehyde, an electron microscopy (EM) grade fixative. The cells were then processed according to previously established protocols [\[57\]](#). Following this, the samples were examined using a JEM-F200 transmission electron microscope, allowing for lipid droplet imaging and detailed visualization of the cellular structures.

### ***In-vitro* cholesterol trafficking assay**

KCs were first treated with NBD-cholesterol for 16Hrs, allowing the fluorescent cholesterol analog to integrate into their membranes. Next, KCs were washed with RPMI 1640 and cultured for again 2Hrs in the same medium to eliminate any residual fluorescence from the background. These labelled KCs were infected with LD for 4Hrs and washed to remove loosely attached MFs. The KCs were then either immediately fixed with 2% paraformaldehyde (PFA) or left untreated to serve as a control group. Finally, the fixed MFs were examined under a confocal microscope.

### **Flow cytometry to determine NBD-cholesterol**

KCs were first treated with NBD-cholesterol as previously described. LD parasites were isolated after 2Hrs of attachment to the KCs. The quenching of NBD-cholesterol was measured by flow cytometric analysis, using an unstained control.

### **Total internal reflection fluorescence (TIRF) microscopy**

To visualize membrane fluidity, TIRF microscopy was performed [75]. Briefly, KCs were plated on coverslips, and after 24Hrs of LD-infection, KCs were fixed with 2% PFA. KCs were stained with Laurdan dye and visualized with Olympus cell TIRF microscopic system. Representative images of three independent biological experiments have been provided.

### **RNAseq analysis**

RNA-Sequencing was performed utilizing external service by Bionivid Project for LD-S and LD-R-infected-PECs, following 24Hrs of infection keeping uninfected MΦs as control (Bioproject. Average expression of differentially expressed genes related to lipid metabolism between LD-S and LD-R-infected-KC is represented as heat map. RAW sequencing reads have been deposited to Indian Nucleotide Data Archive (<https://india.rcb.ac.in/home>) with accession number INRP000146.

### **Lipid profiling from serum**

Lipid profile analysis of patient serum was conducted using lipid estimation kits employing the enzymatic method for the determination of cholesterol, triglycerides, HDL, and LDL. For BALB/c mice, blood was collected following anesthesia and prior to euthanasia. The blood samples were maintained at 4°C to allow clotting, after which the serum was isolated.

### **Evaluation of EC50**

KCs infected with LD parasites were treated with SAG or Amp-B 24Hrs post-infection and incubated for an additional 24Hrs. At the experimental endpoints, coverslips were washed with PBS, air-dried, and fixed with 100% ice-cold methanol. Subsequently, the coverslips were stained with Giemsa solution and examined microscopically. To quantify the number of amastigotes, one hundred KCs per coverslip were counted. The average of three untreated coverslips was taken as 100% control, and the percentage inhibition of infected KCs in treated cultures was calculated. EC50 values for each isolate were estimated against each drug. Results were also expressed as EC50.

### **Aspirin treatment and amastigote load determination**

KCs were infected with LD parasites and, after 4Hrs, treated with 5μM Aspirin for a duration of 24 hours. Following this treatment, the KCs were washed and subsequently treated with Amp-B at a specific dose for an additional 24 hours. Coverslips were then washed with PBS, air-dried, and fixed with 100% ice-cold methanol. After fixation, the coverslips were stained with Giemsa solution and examined microscopically. To quantify the number of amastigotes, one hundred KCs per coverslip were counted.

### siRNA mediated Knock down (KD)

For all siRNA transfections, Lipofectamine 2000 was used according to the manufacturer's instructions. PECs were seeded into 24-well plates at a density of  $1 \times 10^5$  per well, and incubated at 37°C with 5% CO<sub>2</sub>. The transfection complex, comprising 1 µl Lipofectamine 2000 (Invitrogen) and specific siRNAs, was directly added to the incubated PECs. Gene silencing was checked by IFA as mentioned previously.

### Statistical analysis

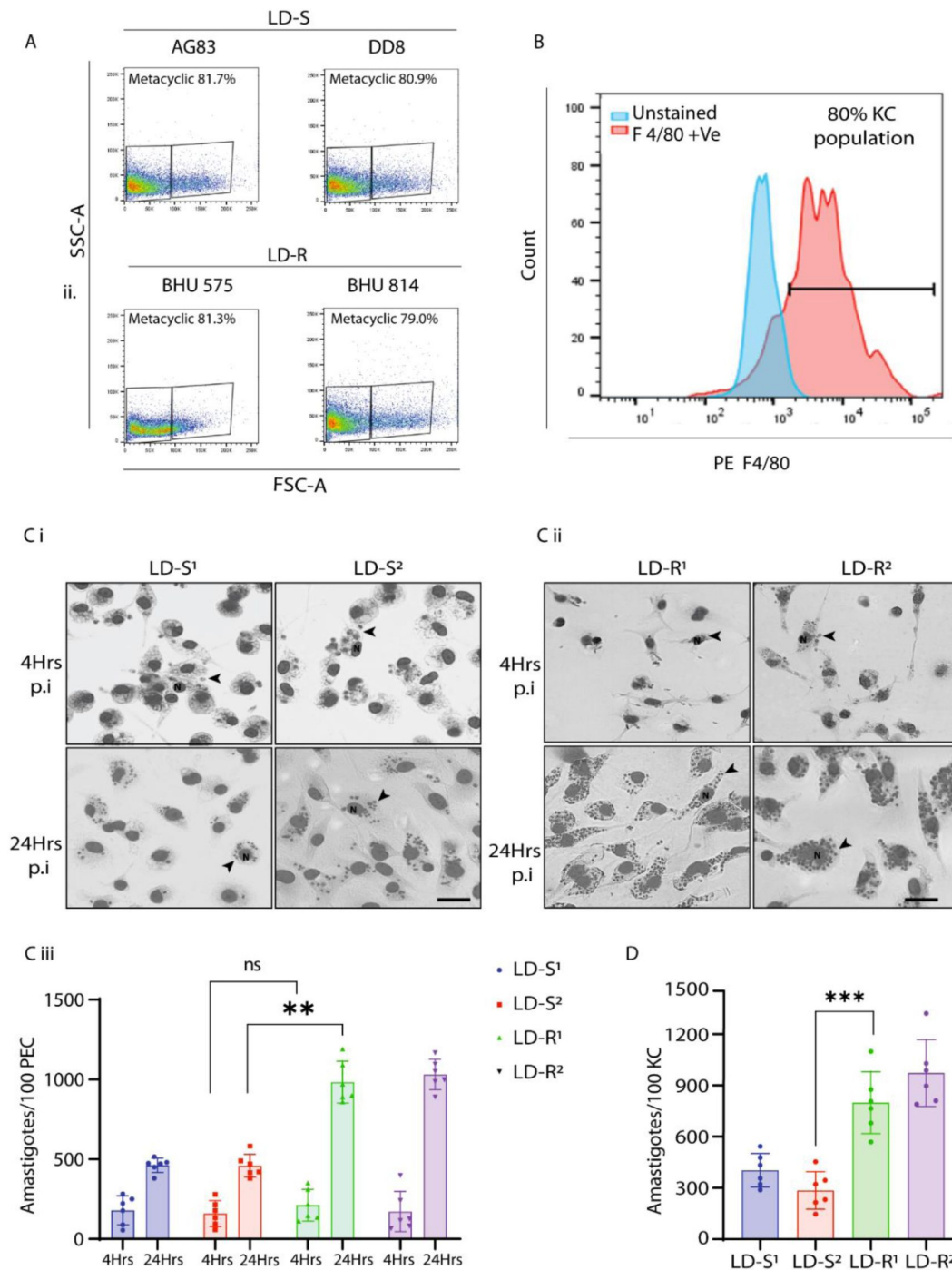
All statistical analyses were performed using GraphPad Prism on raw data. For datasets involving single parameters or single data groups, one-way or two-way analysis of variance (ANOVA) was utilized, followed by Tukey's test with 95% confidence interval. Wilcoxon rank sum analyses, with 95% confidence interval, were employed for comparing multiple groups. To check if the difference between two groups is statistically significant, we used Student t-test. The data are represented as mean  $\pm$  SEM. Statistical significance was determined at  $P \leq 0.05$ . Notation for significance levels: \* $P \leq 0.05$ ; \*\* $P \leq 0.001$ ; \*\*\* $P \leq 0.0001$ .

### Data availability

RAW sequencing reads have been deposited to Indian Nucleotide Data Archive (<https://inda.rcb.ac.in/home> ) with accession number INRP000146.

### Supplementary figures

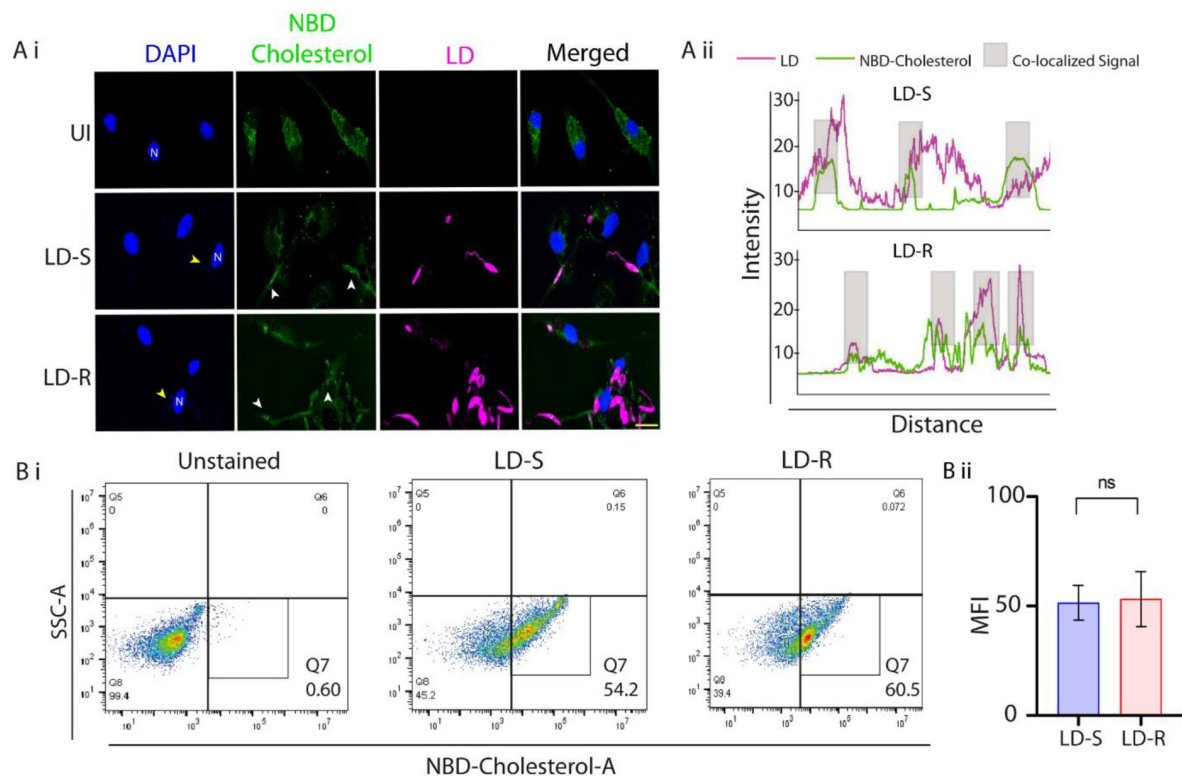




**Figure S1.**

### Kupffer cells (KC) as an *in vitro* model for LD-infection.

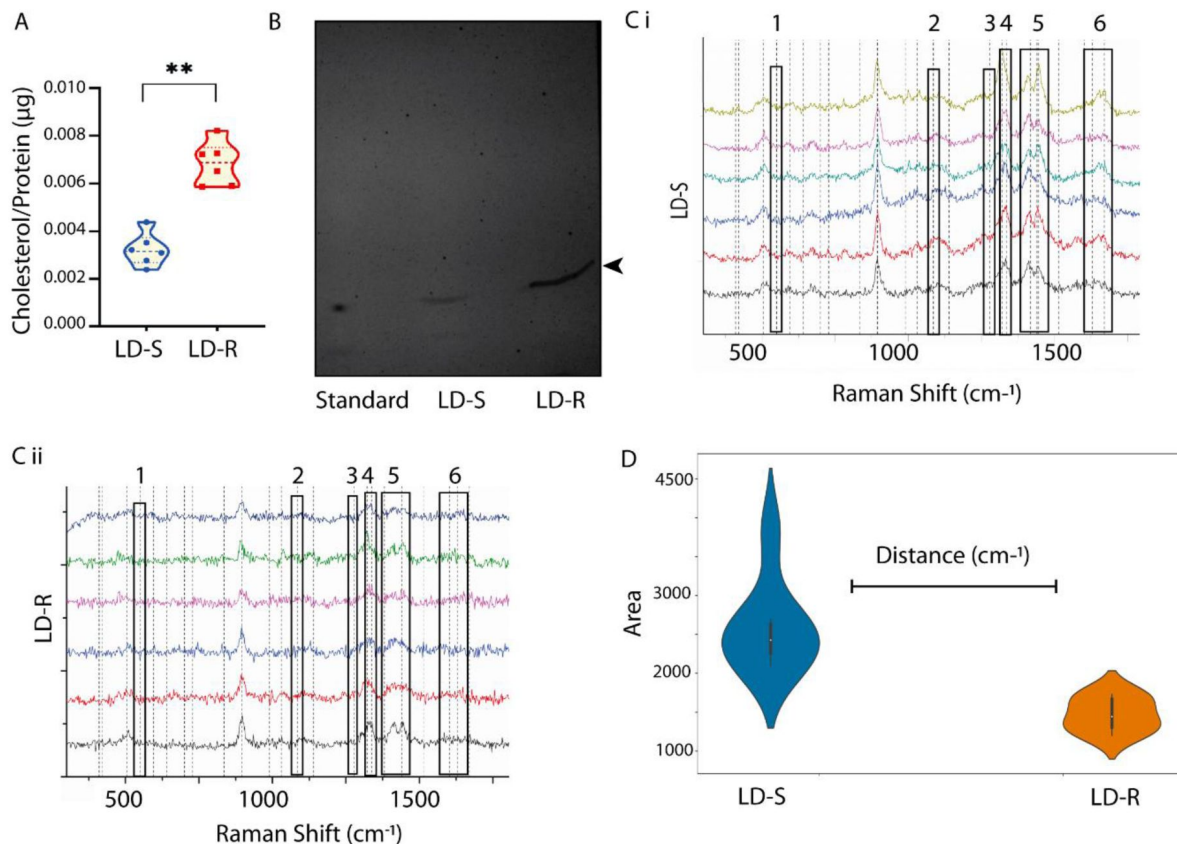
(A) Flow Cytometry representing % Metacyclics LD: (i) LD-S (AG83, DD8) and (ii) LD-R (BHU814, BHU575) based on FSC and SSC. Left, small box with low FSC and SSC gate for metacyclic parasites and right panel with high FSC and SSC gated procyclic parasites. (B) Flow cytometry of F4/80-PE positive KC from the non-parenchymal cell population isolate from murine liver. C Representative Giemsa-stained images LD-infected-PEC, 4 and 24Hrs p.i for (i) LD-S and (ii) LD-R strains. Giemsa Images are represented in grey scale to clearly represent LD nucleus (black arrow). Scale Bar 20µM C (iii) Intracellular amastigote count for 4Hrs and 24Hrs p.i. Each dot represent count from 100 infected- KC. N=6. D Amastigote count from Giemsa-stained image of KC-infected with LD amastigotes, isolated from the spleen of 28 days LD-infected-mice. Each dot represent mean from 100 infected- KCs. N=6. \*\*\* signifies p value < 0.0001, \*\* signifies p value < 0.001, n.s non-significant.



**Figure S2.**

### LD-S and LD-R quenches host membrane cholesterol during initial entry. A

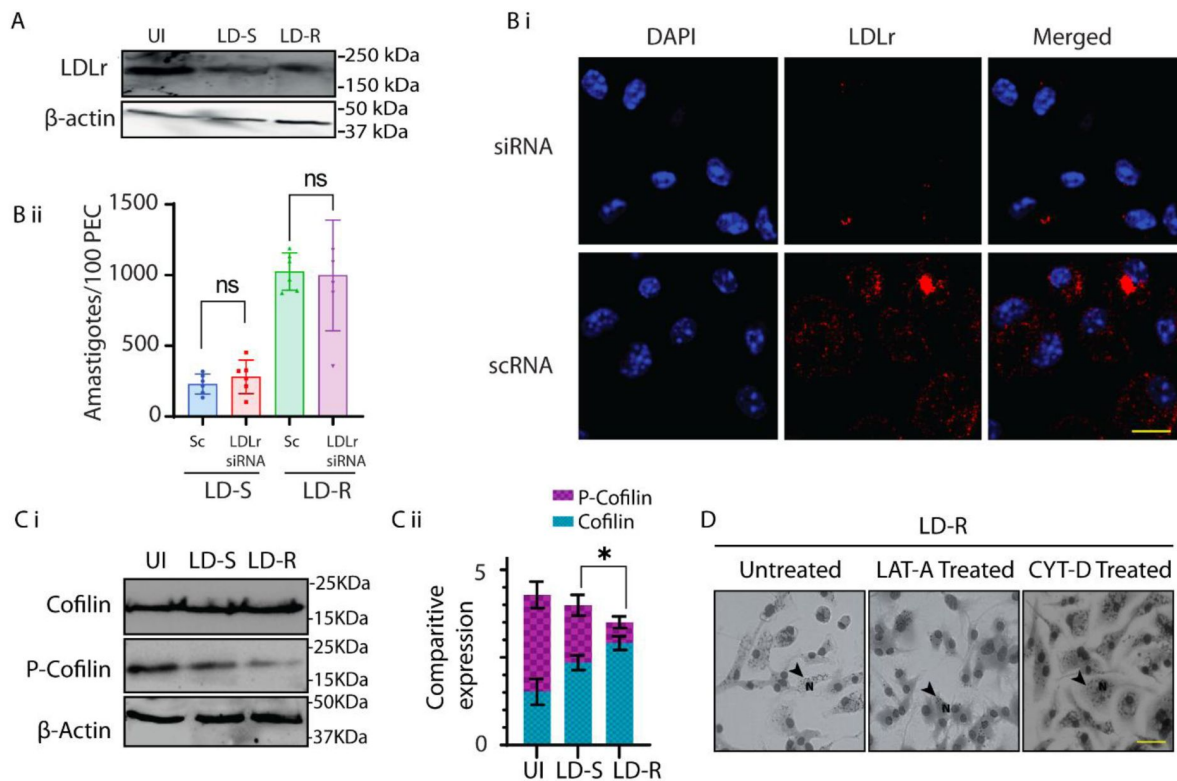
**(i)** KCs were pre-incubated with NBD-cholesterol for 16Hrs. Subsequently, KCs were infected with RFP expressing LD-S or LD-R for 4Hrs or left uninfected and imaged. NBD-cholesterol was excited at 488nm, and LDs are presented with magenta pseudo color. White arrow indicates the NBD- cholesterol within LD while yellow arrow represents LD nucleus. Scale bar 5µm. **A (ii)** Co- localization analysis of LD (magenta), and NBD-Cholesterol (green) from A (i), marked in grey transparent box. **B (i)** Representative Flow cytometry-based analysis of NBD in LD-amastigotes isolated from LD-infected-KCs 2Hrs after infection pre-incubated with NBD cholesterol. Black Box representing % NBD positive population. **B (ii)** Graphical representation of MFI for three independent experiments as in B (i). n.s non-significant.



**Figure S3.**

### LD-R-PV have higher accumulation of Cholesterol and Fatty acids.

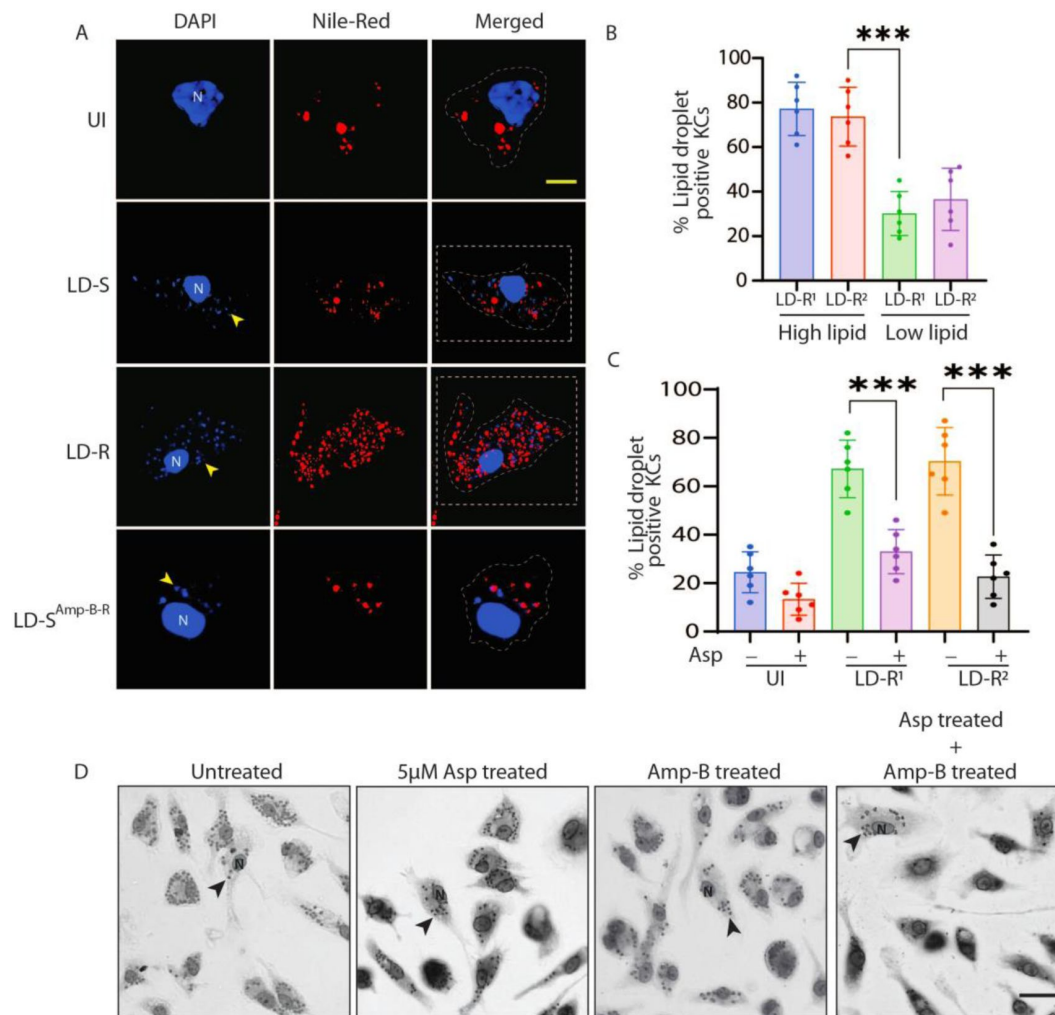
(A) Detection of PV Cholesterol, isolated from LD-S or LD-R-infected KC. Data represented as μg of protein/ μg of Cholesterol, (N=6). (B) TLC profile of isolated lipids from 24Hrs LD-infected-PVs with Cholesterol standard kept as control. (C) Individual Raman spectra from six independent LD- infected-KCs (i) LD-S-infected and (ii) LD-R-infected. Lipid-related peaks are demarcated within the spectra, including those at 1.540-560 cm⁻¹ (Cholesterol), 2. 1080-1090 cm⁻¹ (phospholipids), 3. 1270-1280 cm⁻¹ (triglycerides), 4. 1300-1340 cm⁻¹ (Amide-(II) I bond), 5. 1440-1453 cm⁻¹ (Fatty acids and Triglycerides), and 6. 1650-1660 cm⁻¹ (Amide-I bond). (D) Area under the curve, and shift are calculated and plotted for each experimental condition as represented in C (i) and (ii). \*\* signifies p value <0.001,



**Figure S4.**

#### Latrunculin-A restricts extracellular LDL influx inhibiting LD-R amastigote proliferation.

**(A)** Western blot comparing of LDLr expression between uninfected, LD-S, and LD- R-infected-KC 24Hrs p.i. **B (i)** Expression of LDLr determined by immunofluorescence in PECs treated with LDLr siRNA and scrambled siRNA (Sc). Scale bar 10µm. **B (ii)** Graph representing number of LD-amastigotes in LDLr knockdown per 100 PECs for B (i) (N = 6). Data: mean ± SD. **C (i)** Cofilin and phosphorylated-Cofilin expression by Western blot in LD-infected-PEC. **C (ii)** Graphical representation of comparative expression of abundance of Cofilin and phosphorylated- Cofilin with respect to β-actin for C(i). **(D)** Giemsa stained images of Latrunculin-A (LAT-A) and Cytochalasin-D (CYT-D) treated LD-infected-KCs in grey scale. LD nucleus are represented with Black arrow and host nucleus are represented as N. Scale bar 20µm. n.s non-significant.



**Figure S5.**

**Aspirin treatment inhibits Lipid droplets formation and increase responsiveness of LD-R amastigotes towards Amphotericin-B.**

**(A)** Determination of lipid droplet accumulation by Nile red staining in LD-S and LD-R-infected-KC or KC left uninfected. Dotted line mark the cell periphery while dotted box cells were used for 3D reconstruction and represent in Movie 3. Scale Bar 5 µm. **(B)** Graphical representation of Lipid Droplet positive KCs in high lipid and low lipid media. Data represented as % of lipid droplet positive KCs (N=6) collected from 3 independent experiments. **(C)** Effect of Aspirin(5µM) on Lipid droplets accumulation in LD-R-infected-KCs. Independent infection was performed with LD-R<sup>1</sup> and LD-R<sup>2</sup> strains (N=6) and % KCs with high lipid droplets accumulation are represented. **(D)** Representative Giemsa stained images of LD-R- infected-KCs in different experimental conditions either untreated or treated with Aspirin (5µM) or Amp-B (0.36 µM) or with a combination of Aspirin (5µM) and Amp-B (0.11 µM) at 48Hrs p.i. Scale bar 20µm. \*\*\* signifies p value < 0.0001



## Acknowledgements

SP acknowledges CSIR-UGC (2020-2021) and PMRF (2021-2024) fellowship. DD is a recipient of GATE fellowship. DM is a recipient of CSIR fellowship, SC is a recipient of GATE fellowship. Authors would like to acknowledge IIR lab, SMST, IITKGP for their help in kupffer cell isolation and Flow cytometric analysis, Confocal Microscope facility under DST-FIST grant conferred on the School of Bioscience, File no. SR/FST/LS-I/2019/595(C), and Confocal facility of Department of Biotechnology, DST-FIST, Govt. of India. Authors would like to thank Prof. Syamal Roy, IICB, India for providing all the LD strains used in this study and Prof. Yasuyuki Goto, The University of Tokyo, Japan for allowing experiments related to live cell video microscopy in his lab. Authors acknowledges CRF SMST, and CRF IITKGP. Authors would like to thanks Dr. Moumita Bhaumik, NICED, India for helping in membrane isolation experiments. Authors would like to thank all the interns of the IDI lab, SMST who help in various experiments related to this project.

## Additional information

### Author Contributions

SP conceptualization, experiments, analysis, MS preparation. DD performed Raman spectroscopy experiments. DM generated all the GFP, RFP expressing LD strains, SC perform experiments related to EC50, AB performed Raman spectroscopy experiments and analysis, KC did inhibitor study related to Cyt-D, Lat-A, RM serum collection from VL patients, AS related serum lipid profile, formal analysis of the MS, KP related to serum lipid profile, formal analysis of the MS, SD helped in formal analysis of Raman spectroscopic experiments. B.M conceptualization, analysis, conceived and directed the project and prepared the MS with input from all the authors.

## Additional files

**Movies 1–3.** [🔗](#)

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## Reviewer #1 (Public review):

Although the use of antimony has been discontinued in India, the observation that there are *Leishmania* parasites that are resistant to antimony in circulation has been cited as evidence that these resistant parasites are now a distinct strain with properties that ensure their transmission and persistence. It is of interest to determine what are the properties that favor the retention of their drug resistance phenotype even in the absence of the selective pressure that would otherwise be conferred by the drug. The hypothesis that these authors set out to test is that these parasites have developed a new capacity to acquire and utilize lipids, especially cholesterol which affords them the capacity to grow robustly in infected hosts.

### Major issues:

There are several experiments for which they do not provide sufficient details, but proceed to make significant conclusions.

Experiments in section 5 are poorly described. They supposedly isolated PVs from infected cells. No details of their protocol for the isolation of PVs are provided. They reference a protocol for PV isolation that focused on the isolation of PVs after *L. amazonensis* infection. In the images of infection that they show, by 24 hrs, infected cells harbor a considerable number of parasites. Is it at the 24 hr time point that they recover PVs? What is the purity of PVs? The authors should provide evidence of the success of this protocol in their hands. Earlier, they mentioned that using imaging techniques, the PVs seem to have fused or interconnected somehow. Does this affect the capacity to recover PVs? If more membranes are recovered in the PV fraction, it may explain the higher cholesterol content.

In section 6 they evaluate the mechanism of LDL uptake in macrophages. Several approaches and endocytic pathway inhibitors are employed. The authors must be aware that the role of cytochalasin D in the disruption of fluid phase endocytosis is controversial. Although they reference a study that suggests that cytochalasin D has no effect on fluid-phase endocytosis, other studies have found the opposite (doi: 10.1371/journal.pone.0058054). It wasn't readily evident what concentrations were used in their study. They should consider testing more than 1 concentration of the drug before they make their conclusions on their findings on fluid phase endocytosis.

In Figure 5 they present a blot that shows increased Lamp1 expression from as early as 4 hrs after infection with LD-R and by 12 hrs after infection of both LD-S and LD-R. Increased Lamp1 expression after *Leishmania* infection has not been reported by others. By what mechanism do they suggest is causing such a rapid increase (at 4hrs post-infection) in Lamp-1 protein? As they report, their RNA seq data did not show an increase in LAMP1 transcription (lines 432 - 434).

In Figure 6, amongst several assays, they reported on studies where SPC-1 is knocked down in PECs. They failed to provide any evidence of the success of the knockdown, but nonetheless showed greater LD-R after NPC-1 was knocked down. They should provide more details of such experiments.

### Minor issues

There is an implication that parasite replication occurs well before 24hrs post-infection? Studies on *Leishmania* parasite replication have reported on the commencement of replication after 24hrs post-infection of macrophages (PMCID: PMC9642900). Is this dramatic increase in parasite numbers that they observed due to early parasite replication?

Several of the fluorescence images in the paper are difficult to see. It would be helpful if a blown-up (higher magnification image of images in Figure 1 (especially D) for example) is

presented.

The times at which they choose to evaluate their infections seem arbitrary. It is not clear why they stopped analysis of their KC infections at 24 hrs. As mentioned above, several studies have shown that this is when intracellular amastigotes start replicating. They should consider extending their analyses to 48 or 72 hrs post-infection. Also, they stop in vitro infection of Apoe<sup>-/-</sup> mice at 11 days. Why? No explanation is given for why only 1 point after infection.

<https://doi.org/10.7554/eLife.102857.1.sa2>

#### **Reviewer #2 (Public review):**

##### **Summary:**

This study by Pradhan et al. offers critical insights into the mechanisms by which antimony-resistant *Leishmania donovani* (LD-R) parasites alter host cell lipid metabolism to facilitate their own growth and, in the process, acquire resistance to amphotericin B therapy. The authors illustrate that LD-R parasites enhance LDL uptake via fluid-phase endocytosis, resulting in the accumulation of neutral lipids in the form of lipid droplets that surround the intracellular amastigotes within the parasitophorous vacuoles (PV) that support their development and contribute to amphotericin B treatment resistance. The evidence provided by the authors supporting the main conclusions is compelling, presenting rigorous controls and multiple complementary approaches. The work represents an important advance in understanding how intracellular parasites can modify host metabolism to support their survival and escape drug treatment.

##### **Strengths:**

- (1) The study utilizes clinical isolates of antimony-resistant *L. donovani* and provides interesting mechanistic information regarding the increased LD-R isolate virulence and emerging amphotericin B resistance.
- (2) The authors have used a comprehensive experimental approach to provide a link between antimony-resistant isolates, lipid metabolism, parasite virulence, and amphotericin B resistance. They have combined the following approaches:
  - (a) In vivo infection models involving BL/6 and Apoe<sup>-/-</sup> mice.
  - (b) Ex-vivo infection models using primary Kupffer cells (KC) and peritoneal exudate macrophages (PEC) as physiologically relevant host cells.
  - (c) Various complementary techniques to ascertain lipid metabolism including GC-MS, Raman spectroscopy, microscopy.
  - (d) Applications of genetic and pharmacological tools to show the uptake and utilization of host lipids by the infected macrophage resident *L. donovani* amastigotes.
- (3) The outcome of this study has clear clinical significance. Additionally, the authors have supported their work by including patient data showing a clear clinical significance and correlation between serum lipid profiles and treatment outcomes.
- (4) The present study effectively connects the basic cellular biology of host-pathogen interactions with clinical observations of drug resistance.
- (5) Major findings in the study are well-supported by the data:
  - (a) Intracellular LD-R parasites induce fluid-phase endocytosis of LDL independent of LDL receptor (LDLr).
  - (b) Enhanced fusion of LDL-containing vesicles with parasitophorous vacuoles (PV) containing LD-R parasites both within infected KCs and PECs cells.
  - (c) Intracellular cholesterol transporter NPC1-mediated cholesterol efflux from

parasitophorous vacuoles is suppressed by the LD-R parasites within infected cells.  
(d) Selective exclusion of inflammatory ox-LDL through MSR1 downregulation.  
(e) Accumulation of neutral lipid droplets contributing to amphotericin B resistance.

Weaknesses:

The weaknesses are minor:

(1) The authors do not show how they ascertain that they have a purified fraction of the PV post-density gradient centrifugation.

(2) The study could have benefited from a more detailed analysis of how lipid droplets physically interfere with amphotericin B access to parasites.

Impact and significance:

This work makes several fundamental advances:

(1) The authors were able to show the link between antimony resistance and enhanced parasite proliferation.

(2) They were also able to reveal how parasites can modify host cell metabolism to support their growth while avoiding inflammation.

(3) They were able to show a certain mechanistic basis for emerging amphotericin B resistance.

(4) They suggest therapeutic strategies combining lipid droplet inhibitors with current drugs.

<https://doi.org/10.7554/eLife.102857.1.sa1>

#### Author response:

##### **eLife Assessment:**

*This important study investigates the propensity of the intravacuolar pathogen, Leishmania, to scavenge lipids which it utilizes for its accelerated growth within macrophages. Although some of the data compellingly links increased lipid acquisition to parasite growth, data to support the underlying mechanism to describe the proposed model is incomplete. The study adds to other work that has implicated pathogen-derived processes in the selective recruitment of vesicles to the pathogen-containing vacuole, based on the content of the cargo.*

We appreciate the time and effort that Editor and Reviewers have provided to provide the assessment of our work (eLife: eLife-RP-RA-2024-102857). We thank them all for this assessment.

Regarding some of the concerns raised by Reviewer 1, particularly the lack of data on NPC-1 knockdown, we would like to clarify that this information was included in our original submission (as elaborated in detail in the following section). Additionally, we acknowledge that one of the major concerns about the completeness of our work stems from Reviewer 1's comments on the isolation and purity of the parasitophorous vacuole (PV). Reviewer 2 has also emphasized the importance of this experiment in strengthening the technical rigor of our study, and we fully agree with this recommendation. We acknowledge that this is a very appropriate suggestion by both the Reviewers and we will include this data in the subsequent revision of this work for revaluation of assessment. Also, ahead of a full revision of the paper, we would like to address the concerns raised by the reviewers outlining our revision plans.



## Public Reviews:

### Reviewer #1 (Public review):

*Although the use of antimony has been discontinued in India, the observation that there are Leishmania parasites that are resistant to antimony in circulation has been cited as evidence that these resistant parasites are now a distinct strain with properties that ensure their transmission and persistence. It is of interest to determine what are the properties that favor the retention of their drug resistance phenotype even in the absence of the selective pressure that would otherwise be conferred by the drug. The hypothesis that these authors set out to test is that these parasites have developed a new capacity to acquire and utilize lipids, especially cholesterol which affords them the capacity to grow robustly in infected hosts.*

We sincerely appreciate Reviewer 1's thoughtful and positive evaluation of our manuscript. We acknowledge that the reviewer has a few major concerns, and we would like to address them one by one in the following section of this initial response before submitting a full revision of our work.

### Major issues:

*(1) There are several experiments for which they do not provide sufficient details, but proceed to make significant conclusions.*

*Experiments in section 5 are poorly described. They supposedly isolated PVs from infected cells. No details of their protocol for the isolation of PVs are provided. They reference a protocol for PV isolation that focused on the isolation of PVs after L. amazonensis infection. In the images of infection that they show, by 24 hrs, infected cells harbor a considerable number of parasites. Is it at the 24 hr time point that they recover PVs? What is the purity of PVs? The authors should provide evidence of the success of this protocol in their hands. Earlier, they mentioned that using imaging techniques, the PVs seem to have fused or interconnected somehow. Does this affect the capacity to recover PVs? If more membranes are recovered in the PV fraction, it may explain the higher cholesterol content.*

We would like to thank the reviewer for correctly pointing out lack of details regarding PV isolation and its purity. There are multiple questions raised by the reviewer and we will answer them one by one in a point wise manner:

Firstly, “Is it at the 24 hr time point that they recover PVs?”

In the ‘Methods’ section of the original submission (Line number-606-611), there is a separate section on “Parasitophorous vacuole (PV) Isolation and cholesterol measurement”, where it is clearly mentioned, “24Hrs LD infected KCs were lysed by passing through a 22-gauge syringe needle to release cellular contents. Parasitophorous vacuoles (PV) were then isolated using a previously outlined protocol [Ref: 73].” However, we do acknowledge further details might be useful to enrich this section, and hence we would like to include the following details in the revised manuscript, “ $10^7$  KCs were seeded in a 100 mm plate and allowed to adhere for 24 hours. Following infection with Leishmania donovani (LD) for 24 hours, the infected KCs were harvested by gentle scraping and lysed through five successive passages through an insulin needle to ensure membrane disruption while preserving organelle integrity. The lysate was centrifuged at  $200 \times g$  for 10 minutes at  $4^\circ\text{C}$  to remove intact cells and large debris. The resulting supernatant was carefully collected and subjected to a discontinuous sucrose density gradient (60%, 40%, and 20%). The gradient was centrifuged at  $700 \times g$  for 25 minutes at  $4^\circ\text{C}$  to facilitate organelle separation. The interphase between the 40% and 60% sucrose

layers, enriched with PVs, was carefully collected and subjected to a final centrifugation step at  $12,000 \times g$  for 25 minutes at  $4^{\circ}\text{C}$ . The supernatant was discarded, and the resulting pellet was enriched for purified parasitophorous vacuoles, suitable for downstream biochemical and molecular analyses.”

Secondly, What is the purity of PVs? Earlier, they mentioned that using imaging techniques, the PVs seem to have fused or interconnected somehow. Does this affect the capacity to recover PVs? If more membranes are recovered in the PV fraction, it may explain the higher cholesterol content.

We appreciate the reviewer for pointing this critical lack of data in the current version of the manuscript. We will be providing data on the purity of isolated fraction by performing western blot against PV and cytoplasmic fraction in the Revised manuscript. We admit, as rightly pointed out by the reviewer we need to access the purity of isolated PV in our experiment and we plan to show this in the Revised manuscript along with a biochemical quantification of total PV membrane isolated under different experimental condition using Amplex Red kit (Invitrogen™ A12216) or similar other methods.

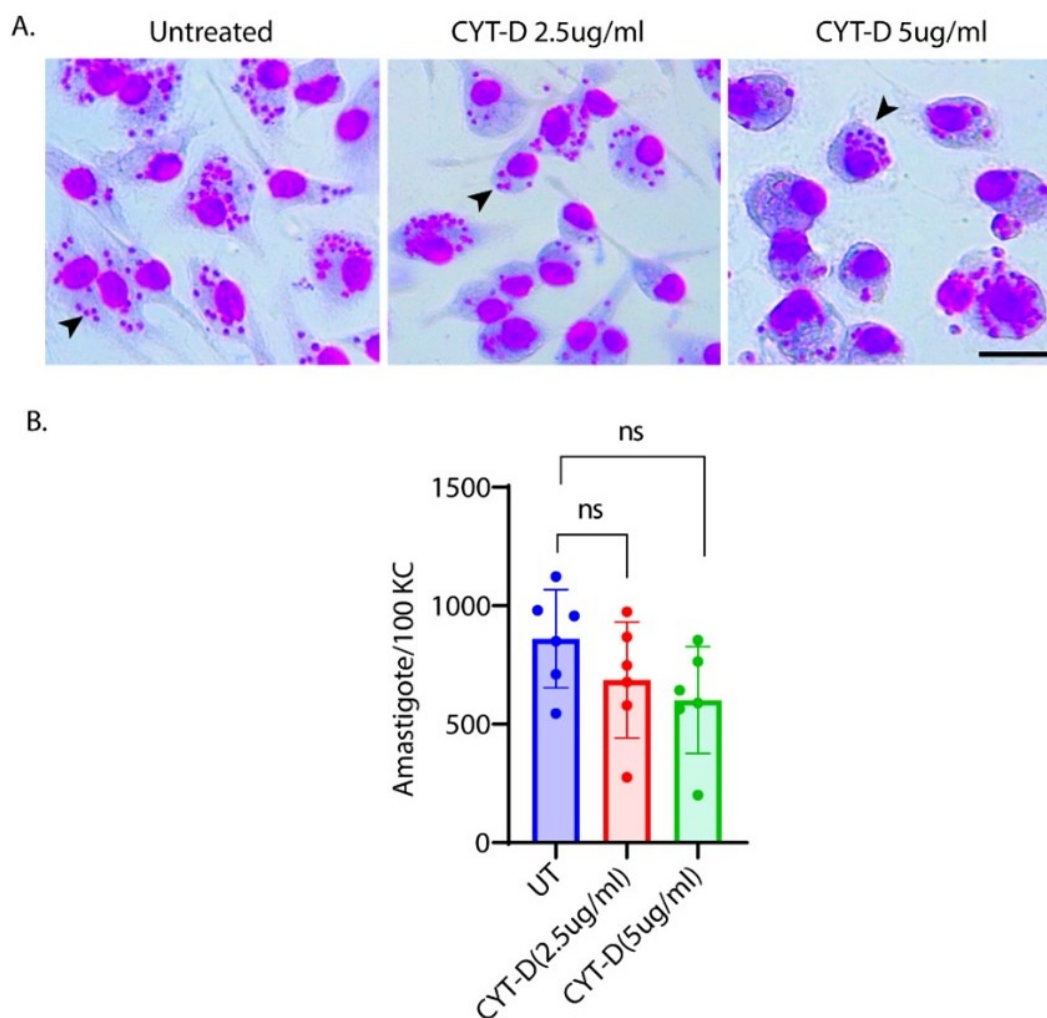
*(2) In section 6 they evaluate the mechanism of LDL uptake in macrophages. Several approaches and endocytic pathway inhibitors are employed. The authors must be aware that the role of cytochalasin D in the disruption of fluid phase endocytosis is controversial. Although they reference a study that suggests that cytochalasin D has no effect on fluid-phase endocytosis, other studies have found the opposite (doi: 10.1371/journal.pone.0058054). It wasn't readily evident what concentrations were used in their study. They should consider testing more than 1 concentration of the drug before they make their conclusions on their findings on fluid phase endocytosis.*

We thank the reviewer for this insightful comment and we apologise for missing out mentioning Cytochalasin D concentration. To clarify, LDL uptake by LD-R infected KCs is LDL-receptor independent as clearly shown in Section 6, Figure 4A, Figure S4A, Figure S4B i and Figure S4B ii in the Submitted manuscript. In (Figure 4F and Figure S4D) of the Submitted manuscript, as referred by the Reviewer, Cytochalasin D was used at a concentration of  $2.5\mu\text{g/ml}$ . At this concentration, we did not observe any effect of Cytochalasin D on LDL-receptor independent fluid phase endocytosis as intracellular LD-R amastigotes was able to uptake LDL successfully and proliferate in infected Kupffer cells, unlike Latrunculin-A ( $5\mu\text{M}$ ) treatment which completely inhibited intracellular proliferation of LD-R amastigotes by blocking only receptor independent Fluid phase endocytosis (Movie 2A and 2B and Figure 4E in the Submitted manuscript). In fact, the study referred by the reviewer (doi: 10.1371/journal.pone.0058054), used a concentration of  $4\mu\text{g/ml}$  Cytochalasin D which did affect both LDL-receptor dependent and also receptor independent endocytosis in bone marrow derived macrophages. We would also like to clarify that in this work during our preliminary experiments we have also tested higher concentration Cytochalasin-D ( $5\mu\text{g/ml}$ ). However, even at this higher concentration there were no significant effect of Cytochalasin-D on LD-R induced LDL-receptor independent fluid phase endocytosis as observed from intracellular LD-R amastigote count represented in Author response image 1. Thus, we strongly believe that Cytochalasin D does not have any impact on LD-R induced fluid phase endocytosis even at higher concentration. We will include this in the discussion section of the revised manuscript to clear out any confusion that readers might have, and also concentration of all the inhibitors used in the study will be mentioned in the Result section, as well as in the revised Figure legends.

#### Author response image 1.

A. Giemsa-stained images illustrating the impact of concentrations of CYT-D ( $2.5$  and  $5\mu\text{g/ml}$ ) on LD-R-infected Kupffer cells. Black arrow showing intracellular amastigotes. Scale bar

10 $\mu$ M. B. Graphical representation depicting the effect of varying concentrations of CYT-D on the intracellular growth of LD-R. 'ns' depicts no significant change.



(3) In Figure 5 they present a blot that shows increased Lamp1 expression from as early as 4 hrs after infection with LD-R and by 12 hrs after infection of both LD-S and LD-R. Increased Lamp1 expression after Leishmania infection has not been reported by others. By what mechanism do they suggest is causing such a rapid increase (at 4hrs post-infection) in Lamp-1 protein? As they report, their RNA seq data did not show an increase in LAMP1 transcription (lines 432 - 434).

We would like to express our gratitude to the reviewer for highlighting the novelty of this observation. Indeed, to the best of our knowledge, no similar findings have been reported previously in primary macrophages infected with Leishmania donovani (LD). Firstly, we would like to point out, as stated in the Methods section (lines 562–566) of the Submitted manuscript: "Flow-sorted metacyclic LD promastigotes were used at a MOI of 1:10 (with variations of 1:5 and 1:20 in some cases) for 4 hours, which was considered the 0th point of infection. Macrophages were subsequently washed to remove any extracellular loosely attached parasites and incubated further as per experimental requirements." This indicates that our actual study points correspond to approximately the 8th hour post-infection". We just wanted to clarify this to prevent any potential confusion.

Now regarding LAMP1 expression, although we could not find any previous reports of its expression in LD infected primary macrophages, we would like to mention that a previous report ([doi.org/10.1128/mBio.01464-20](https://doi.org/10.1128/mBio.01464-20)), has shown a similar punctuated LAMP-1 upregulation (as observed by us in Figure 5A i of the Submitted manuscript) in response to leishmania infection in non-phagocytic fibroblast. It is tempting to speculate that increased LAMP-1 expression observed in response to LD-R infected macrophages might be due to increased lysosomal biogenesis, required for degrading increased endocytosed-LDL into bioavailable cholesterol. However, since no change in LAMP-1 expression in RNA seq data (Figure 6, of the Submitted manuscript), we can only speculate that this is happening due to some post transcriptional or post translational modifications. But further work will definitely require to investigate this mechanism in details which is beyond the scope of this work. That is why, in the Submitted manuscript, (Line 432-435), we have discussed this, “Although available RNAseq analysis (Figure 6) did not support this increased expression of lamp-1 in the transcript level, it did reflect a notable upregulation of vesicular fusion protein (VSP) vamp8 and stx1a in response to LD-R-infection. LD infection can regulate LAMP-1 expression, and the role of VSPs in LDL-vesicle fusion with LD-R-PV is worthy of further investigation.”

However, we agree with the reviewer that this might not be enough for the clarification. Hence in the revised manuscript we plan to update this part as follows, “Although available RNAseq analysis (Figure 6) did not support this increased expression of lamp-1 in the transcript level, it did reflect a notable upregulation of vesicular fusion protein (VSP) vamp8 and stx1a in response to LD-R-infection. How, LD infection can regulate LAMP-1 expression, and the role of VSPs in LDL-vesicle fusion with LD-R-PV is worthy of further investigation. It is possible and has been earlier reported that LD infection can regulate host proteins expression through post transcriptional and post translational modifications ([doi.org/10.1111/pim.12156](https://doi.org/10.1111/pim.12156), [doi.org/10.3389/fmicb.2017.00314](https://doi.org/10.3389/fmicb.2017.00314), [doi.org/10.3389/fimmu.2023.1287539](https://doi.org/10.3389/fimmu.2023.1287539)). It is tempting to speculate that LD-R amastigote might be promoting an increased lysosomal biogenesis through any such mechanism to increase supply of bioavailable cholesterol through action of lysosomal acid hydrolases on LDL.”

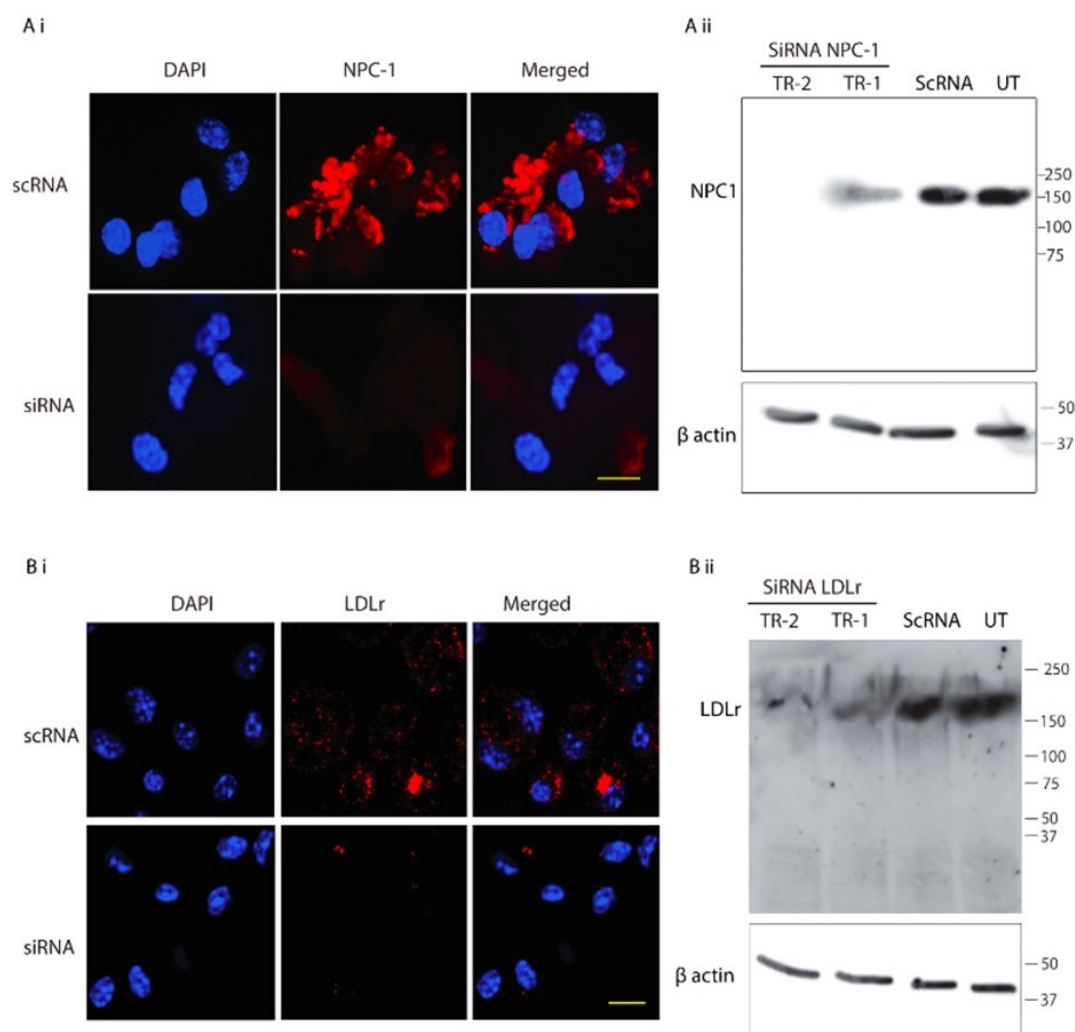
*(4) In Figure 6, amongst several assays, they reported on studies where NPC-1 is knocked down in PECs. They failed to provide any evidence of the success of the knockdown, but nonetheless showed greater LD-R after NPC-1 was knocked down. They should provide more details of such experiments.*

Although we do understand the concern raised by the reviewer, this statement in question is factually incorrect. We would like to point out that in Figure 6 F i, of the Submitted manuscript, we have demonstrated decreased NPC-1 staining following transfection with NPC-1-specific siRNA, whereas no such reduction was observed with scrambled RNA. Similar immunofluorescence data confirming LDL-receptor knockdown has also been provided in Figure S4B i of the Submitted manuscript. However, we acknowledge that the reviewer may be referring to the lack of quantitative validation of the knockdown via Western blot. We would like to clarify although, we already had this data, but we did not include it to avoid duplication to reduce the data density of the manuscript. But as suggested by the reviewer, we will be including western blot for both NPC-1 and LDL-receptor knock down in the revised manuscript as represented in Author response image 2. Additionally, as suggested by the reviewer, we also noticed lack of details in Methods section of the submitted manuscript, concerning siRNA mediated Knock down (KD). Therefore, we plan to include more details in the revised manuscript, which will read as, “For all siRNA transfections, Lipofectamine® RNAiMAX Reagent (Life Technologies, 13778100) specifically designed for knockdown assays in primary cells was used according to the manufacturer's instructions with slight modification. PECs were seeded into 24-well plates at a density of  $1 \times 10^5$  per well, and incubated at 37°C with 5% CO<sub>2</sub>. The transfection complex, comprising (1µl Lipofectamine® RNAiMAX and 50µl Opti MEM) and (1 µl siRNA and 50µl Opti MEM) mixed together directly

added to the incubated PECs. Gene silencing was checked by IFA and by Western blot as mentioned previously”.

## Author response image 2.

SiRNA-mediated gene knockdown analysis. (A-i, A-ii) Representative immunofluorescence microscopy image and corresponding Western blot analysis demonstrating the knockdown efficiency of NPC1 following SiRNA-mediated gene silencing, scale bar 10µm. (B-i, B-ii) Immunofluorescence image and Western blot confirming LDLr knockdown upon SiRNA treatment. Scrambled RNA (ScRNA) was used as a negative control, while Small Interfering RNA (SiRNA) specifically targeted NPC1 and LDLr transcripts, scale bar 10µm. TR-1 and TR-2 represent independent experimental trials.  $\beta$ -Actin was used as an endogenous loading control for Western blot normalization.



### Minor issues

(1) There is an implication that parasite replication occurs well before 24hrs post-infection? Studies on Leishmania parasite replication have reported on the commencement of replication after 24hrs post-infection of macrophages (PMCID: PMC9642900). Is this dramatic increase in parasite numbers that they observed due to early parasite replication?

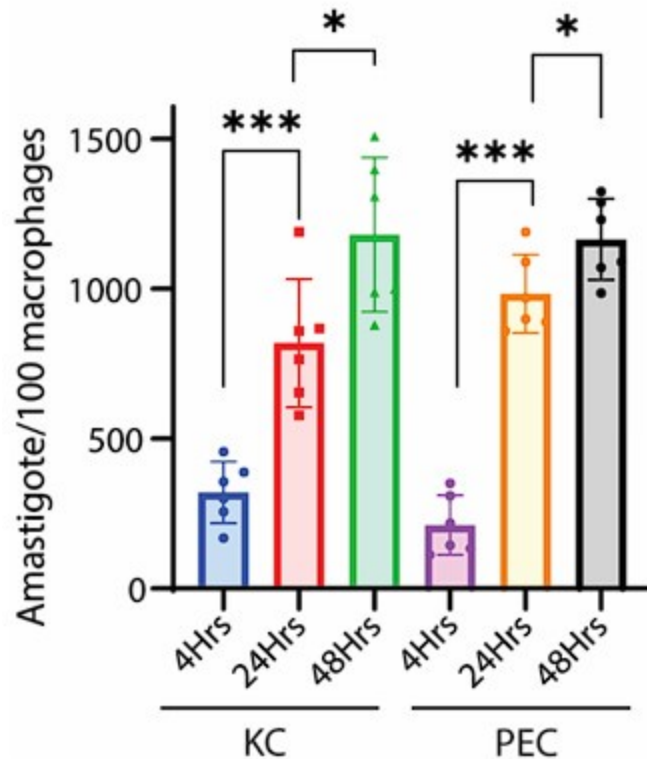


We thank the reviewer for this insightful comment and appreciate the opportunity to clarify our findings. Indeed, as rightly assumed by the Reviewer, as our data suggest, and we also believe that this increase intracellular amastigotes number is a consequence of early replication of *Leishmania donovani*. As already mentioned in response to Point number 3 raised by Reviewer 1, we would again like to highlight that in the Methods section (lines 562–566), it is clearly stated: "Flow-sorted metacyclic LD promastigotes were used at a MOI of 1:10 (with variations of 1:5 and 1:20 in some cases) for 4 hours, which was considered the 0th point of infection. Macrophages were subsequently washed to remove any extracellular loosely attached parasites and incubated further as per experimental requirements." This effectively means that our actual study points correspond to approximately the 8th and 28th hours post-infection and we just want to mention it to avoid any confusion.

Now, regarding specific concern, the study referred by the reviewer on the commencement of replication after 24hrs, was conducted on *Leishmania major*, which may differ significantly from *Leishmania donovani* owing to its species and strain-specific characteristics. In fact, doubling time of *Leishmania donovani* (LD) has been previously reported to be approximately 11.4 hours (doi: 10.1111/j.1550-7408.1990.tb01147.x). Moreover, multiple studies have indicated an exponential increase in intracellular LD amastigote number (more than two-fold increase) by 24hrs post infection. However, by 48hrs post-infection, the replication rate appeared to slow down, with amastigote numbers not increasing (doubling) proportionally (doi:10.1128/AAC.01196-07, doi.org/10.1016/j.ijpara.2011.07.013). We also have a similar observation for both infected PEC and KC as depicted in Figure 1Ci and Figure S1Ci in the Submitted manuscript) along with Author response image 3. Hence it was an informed decision from our side to focus on 24 hours' time point to perform the analysis on intracellular proliferation.

### **Author response image 3.**

Graph representing number of intracellular LD-R (MHOM/IN/2009/BHU575/0) parasite burden at different time points post-infection. \*\*\* signifies p value < 0.0001, \* signifies p value < 0.05.

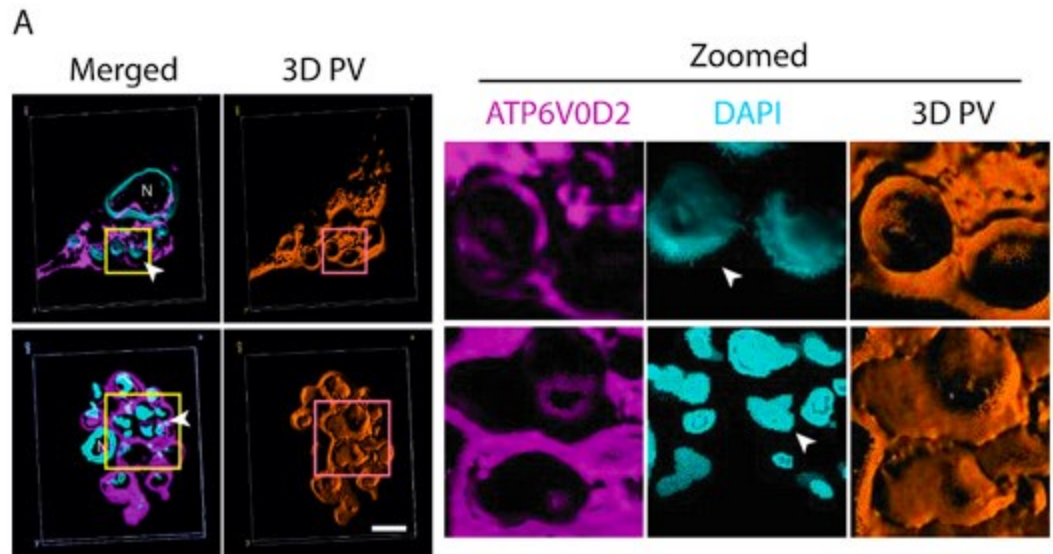


(2) Several of the fluorescence images in the paper are difficult to see. It would be helpful if a blown-up (higher magnification image of images in Figure 1 (especially D) for example) is presented.

We apologise for the inconvenience. Although we have provided Zoomed images for several Figures in the Submitted manuscript, like Figure 4, Figure 5, Figure 6 and Figure 8. However, this was not always doable for all the figures (like for Figure 1D), due to lack of space and Figure arrangements requirements. However, to accommodate Reviewer's request we would like to provide a blown-up image for Figure 1D as represented in Author response image 4 in the Revised version. If the reviewer similar representation for any other particular Figures, we will be happy to perform a similar presentation.

#### Author response image 4.

Three-Dimensional morphometric representation of Parasitophorous Vacuoles (PVs) in *Leishmania* infected Kupffer Cells at 24 Hours Post-Infection: Confocal 3D reconstruction illustrating the spatial distribution of parasitophorous vacuoles (PVs) in Kupffer cells (KCs) infected for 24 hours. ATP6V0D2, a lysosomal vacuolar ATPase subunit, is visualized in magenta, while the nucleus is depicted in cyan. The final panel highlights PV structural grooves outlined in red solid lines, with intracellular *Leishmania donovani* (LD) amastigotes indicated by white arrows. Higher magnification of Figure 1D further emphasizes the increased abundance of PVs in LD-R infected cells, suggesting enhanced intracellular replication and adaptation mechanisms of drug-resistant strains. Scale bar 5µM. Both yellow and magenta solid line box represents the same area of the image.



(3) The times at which they choose to evaluate their infections seem arbitrary. It is not clear why they stopped analysis of their KC infections at 24 hrs. As mentioned above, several studies have shown that this is when intracellular amastigotes start replicating. They should consider extending their analyses to 48 or 72 hrs post-infection. Also, they stop in vitro infection of *Apoe*<sup>-/-</sup> mice at 11 days. Why? No explanation is given for why only 1 point after infection.

Reviewer has raised two independent concerns and we would like to address them individually.

Firstly, “The times at which they choose to evaluate their infections seem arbitrary. It is not clear why they stopped analysis of their KC infections at 24 hrs. As mentioned above, several studies have shown that this is when intracellular amastigotes start replicating. They should consider extending their analyses to 48 or 72 hrs post-infection.”

We have already provided a detail justification for time point selection in our response to Reviewer Minor Comment 1. As mentioned already we observed a significant and sharp rise in the number of intracellular amastigotes between 4 and 24Hrs post-infection (Author response image 4), with replication rate appeared to be not increaseing proportionally after that. This early stage of rapid replication of LD amastigotes, therefore likely coincides with a critical period of lipid acquisition by intracellular amastigotes (Movie 2A and 2B and Figure 4E in the submitted manuscript) and thus 24hrs infected KC was specifically selected. In this regard, we would also like to add that at 72hrs post-infection, we noticed a notable number of infected Kupffer cells began detaching from the wells with extracellular amastigotes probably egressing out from the infected KCs. This phenomenon potentially reflects the severe impact of prolonged infection on Kupffer cell viability and adhesion properties as shown in Author response image 5 and Author Response Video 1. This point further influenced our decision to conclude all infection studies in Kupffer cells by the 48Hrs post-infection, which necessitate to complete the infection time point at 24 Hrs, for allowing treatment of Amp-B for another 24 Hrs (Figure 8, and Figure S5, in the Submitted manuscript). We acknowledge that we should have been possibly more clear on our selection of time points and as the Reviewer have suggested we plan to include this information in the revised manuscript for clear understanding of the reader.

**Author response image 5.**

Representative images of Kupffer cells infected with *Leishmania donovani* at 72Hrs post-infection showing a significant morphological changes. Infected cells exhibit a rounded morphology and progressive detachment. Scale bar 10µm.

72 Hour post infection



Secondly “Also, they stop in vitro infection of *Apoe*<sup>-/-</sup> mice at 11 days. Why? No explanation is given for why only 1 point after infection.”

We apologize for not providing an explanation regarding the selection of the 11-day time point for *Apoe*<sup>-/-</sup> experiments (Figure 2 of the Submitted manuscript). Our rationale for this choice is based on both previous literature and the specific objectives of our study. Previous report suggests that *Leishmania donovani* infection in *Apoe*<sup>-/-</sup> mice triggers a heightened inflammatory response at approximately six weeks’ post-infection compared to C57BL/6 mice, leading to more efficient parasite clearance. This is owing to unique membrane composition of *Apoe*<sup>-/-</sup> which rectifies leishmania mediated defective antigen presentation at a later stage of infection (DOI 10.1194/jlr.M026914). Additionally, previous studies (doi: 10.1128/AAC.47.5.1529-1535.2003) have also indicated that *Leishmania donovani* infection is well-established in vivo within 6 to 11 days post-infection in murine models. Given that in this experiment we particularly aimed to assess the early infection status (parasite load) in diet-induced hypercholesterolemic mice, we would like to argue that the selection of the 11-day time point was intentional and well-aligned with our study objectives as this time point within this window are optimal for capturing initial parasite burden depending on initial lipid utilization, before host-driven immune clearance mechanisms could significantly alter infection dynamics. We will include this explanation in the Revised manuscript as suggested by the Reviewer.

**Reviewer #2 (Public review):**

**Summary:**

*This study by Pradhan et al. offers critical insights into the mechanisms by which antimony-resistant Leishmania donovani (LD-R) parasites alter host cell lipid metabolism to facilitate their own growth and, in the process, acquire resistance to amphotericin B*

*therapy. The authors illustrate that LD-R parasites enhance LDL uptake via fluid-phase endocytosis, resulting in the accumulation of neutral lipids in the form of lipid droplets that surround the intracellular amastigotes within the parasitophorous vacuoles (PV) that support their development and contribute to amphotericin B treatment resistance. The evidence provided by the authors supporting the main conclusions is compelling, presenting rigorous controls and multiple complementary approaches. The work represents an important advance in understanding how intracellular parasites can modify host metabolism to support their survival and escape drug treatment.*

We would like to sincerely thank the reviewer for appreciating our work and find the evidence compelling to address the issue of emergence of drug resistance in infection with intracellular protozoan pathogens. Before we submit a full revision of the paper, we would like to provide a primary response addressing the concerns of the reviewer.

#### *Strengths:*

- (1) The study utilizes clinical isolates of antimony-resistant L. donovani and provides interesting mechanistic information regarding the increased LD-R isolate virulence and emerging amphotericin B resistance.*
- (2) The authors have used a comprehensive experimental approach to provide a link between antimony-resistant isolates, lipid metabolism, parasite virulence, and amphotericin B resistance. They have combined the following approaches:*
  - (a) In vivo infection models involving BL/6 and Apoe<sup>-/-</sup> mice.*
  - (b) Ex-vivo infection models using primary Kupffer cells (KC) and peritoneal exudate macrophages (PEC) as physiologically relevant host cells.*
  - (c) Various complementary techniques to ascertain lipid metabolism including GC-MS, Raman spectroscopy, microscopy.*
  - (d) Applications of genetic and pharmacological tools to show the uptake and utilization of host lipids by the infected macrophage resident L. donovani amastigotes.*
- (3) The outcome of this study has clear clinical significance. Additionally, the authors have supported their work by including patient data showing a clear clinical significance and correlation between serum lipid profiles and treatment outcomes.*
- (4) The present study effectively connects the basic cellular biology of host-pathogen interactions with clinical observations of drug resistance.*
- (5) Major findings in the study are well-supported by the data:*
  - (a) Intracellular LD-R parasites induce fluid-phase endocytosis of LDL independent of LDL receptor (LDLr).*
  - (b) Enhanced fusion of LDL-containing vesicles with parasitophorous vacuoles (PV) containing LD-R parasites both within infected KCs and PECs cells.*
  - (c) Intracellular cholesterol transporter NPC1-mediated cholesterol efflux from parasitophorous vacuoles is suppressed by the LD-R parasites within infected cells.*
  - (d) Selective exclusion of inflammatory ox-LDL through MSR1 downregulation.*
  - (e) Accumulation of neutral lipid droplets contributing to amphotericin B resistance.*

#### *Weaknesses:*



*The weaknesses are minor:*

*(1) The authors do not show how they ascertain that they have a purified fraction of the PV post-density gradient centrifugation.*

*(2) The study could have benefited from a more detailed analysis of how lipid droplets physically interfere with amphotericin B access to parasites.*

*We have addressed both these concerns as our preliminary response in details in subsequent "Recommendations for the Authors section" before we submit a complete Revised manuscript,*

*Impact and significance:*

*This work makes several fundamental advances:*

*(1) The authors were able to show the link between antimony resistance and enhanced parasite proliferation.*

*(2) They were also able to reveal how parasites can modify host cell metabolism to support their growth while avoiding inflammation.*

*(3) They were able to show a certain mechanistic basis for emerging amphotericin B resistance.*

*(4) They suggest therapeutic strategies combining lipid droplet inhibitors with current drugs.*

***Recommendations for the authors:***

***Reviewer #2 (Recommendations for the authors):***

*(1) Experimental suggestions:*

*a) The authors could have provided a more detailed analysis of lipid droplet composition. This is a critically missing piece in this nice study.*

We completely agree with the reviewer on this, a more detailed analysis of lipid droplets composition, dynamics of its formation and mechanism of lipid transfer to amastigotes residing within the PV would be worthy of further investigation. To answer the reviewers, we are already conducting investigation in this direction and have very promising initial results which we are willing to share with the reviewer as unpublished data if requested. Since, we plan to address these questions independently, we hope reviewer will understand our hesitation to include these data into the present work which is already immensely data dense. We sincerely believe existence of lipid droplet contact sites with the PV along with the specific lipid type transfer to amastigotes and its mechanism requires special attention and could stand out as an independent work by itself.

*b) The macrophages (PEC, KC) could have been treated with latex beads as a control, which would indicate that cholesterol and lipids are indeed utilized by the Leishmania parasitophorous vacuole (PV) and essential for its survival and proliferation.*

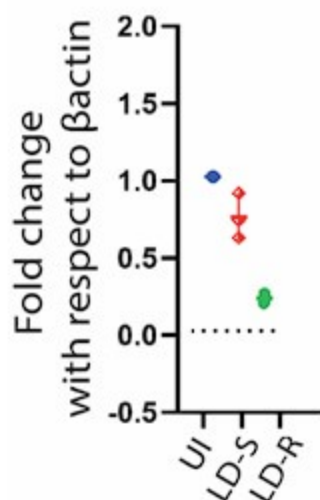
We thank the reviewer for this nice suggestion, which we believe will further strengthen the conclusion of this work. This has also been suggested by Reviewer 1 and we are planning to conduct this experiment and will include this data in the revised version of this manuscript.

c) HMGCoA reductase is an important enzyme for the mevalonate pathway and cholesterol synthesis. The authors have not commented on this enzyme in either host or parasite. Additionally, western blots of these enzymes along with SREBP2 could have been performed.

We appreciate the concern and do see the point why reviewer is suggesting this. We would like to mention that regarding HMGCoA we already do have real time qPCR data which perfectly aligns with our RNAseq data (Figure 6 Ai, in the Submitted manuscript), showing significant downregulation specifically in LD-R infected KC as compared to uninfected control. We are including this data as Author response image 6. However, we did not proceed with checking the level of HMGCoA at the protein level as we noticed several previous reports have suggested that HMGCoA remains under transcriptional control of SERBP2([doi.org/10.1016/j.cmet.2011.03.005](https://doi.org/10.1016/j.cmet.2011.03.005),[doi:10.1194/jlr.C066712](https://doi.org/10.1194/jlr.C066712),[doi:10.1194/jlr.RA119000201](https://doi.org/10.1194/jlr.RA119000201)), which acts the master regulator of mevalonate pathway and cholesterol synthesis ([doi.org/10.1161/ATVBAHA.122.317320](https://doi.org/10.1161/ATVBAHA.122.317320)). However, as suggested by the Reviewer, we will perform this experiment and will update the Revised manuscript with the expression data on HMGCoA probably in the Supplementary section

#### Author response image 6.

qPCR Analysis of HMGCRC Expression Following *Leishmania donovani* Infection: Quantitative PCR analysis showing the relative expression of *hmgcr* (3-hydroxy-3-methylglutaryl-CoA reductase) in Kupffer cells after 24 hours of *Leishmania donovani* (LD) infection compared to uninfected control cells. Gene expression levels are normalized to  $\beta$ -actin as an internal control, and fold change is represented relative to the uninfected condition.



d) The authors should discuss the expression pattern of any enzyme of the mevalonate pathway that they have found to be dysregulated in the transcript data.

As per the reviewer's suggestion, we have already looked into the RNA seq data and observed that apart from *hmgcr*, *hmgcs* (3-hydroxy-3-methylglutaryl-CoA synthase), another key enzyme in the mevalonate pathway, is significantly downregulated in host PECs in response to LD-R infection compared to the LD-S infection. We will update this in the Discussion section of the Revised manuscript, which will read as "Further analysis of RNA sequencing data revealed a

significant downregulation of *hmgcs* (3-hydroxy-3-methylglutaryl-CoA synthase) in LD-R infected PECs as compared to LD-S infection. *HMGCS* which catalyzes the condensation of acetyl-CoA with acetoacetyl-CoA to form 3-hydroxy-3-methylglutaryl-CoA (HMG-CoA), which serves as an intermediate in both cholesterol biosynthesis and ketogenesis. The downregulation of *hmgcs* further supports our observation that LD-R-infected PECs preferentially rely on endocytosed low-density lipoprotein (LDL)-derived cholesterol rather than *de novo* synthesized cholesterol for their metabolic needs.”

*e) The authors have followed a previously published protocol by Real F (reference 73) to enrich for parasitophorous vacuole (PV). However, they do not show how they ascertain that they have a purified fraction of the PV post-density gradient centrifugation. The authors should at least show Western blot data for LAMP1 for different fractions of density gradient from which they enriched the PV.*

As we previously stated in our response to Reviewer 1, the Revised manuscript will include a detailed analysis of purity for different fractions during PV isolation. We sincerely appreciate the reviewer for highlighting this important concern and for suggesting an approach to conduct the experiment. We believe this experiment is crucial and will further reinforce the conclusions of our study.

*(2) Presentation improvements:*

*a) Add a clear timeline for infection experiments.*

Sure. We will be including a schematic of Timelines in the revised figures 2 and 7

*b) Provide more details on patient sample collection and analysis.*

We plan to include more details on the sample collection in the Method section of the Revised manuscript as follows, “Blood samples were collected from a total of 22 individuals spanning a diverse age range (8 to 70 years) by RMRI, Bihar, India. Among these, nine samples were obtained from healthy individuals residing in endemic regions to serve as controls. Serum was isolated from each blood sample through centrifugation, and the lipid profile was subsequently analysed using a specialized diagnostic kit (Coral Clinical System) following the manufacturer’s protocol.”

*c) Consider reorganizing figures to better separate mechanistic and clinical findings.*

We would like to thank the reviewer for this suggestion. However, we feel that the arrangement of the Figures as presented in the Original Submission is really helping a smooth flow of the story and hence, we would not want to disturb that. However, having said that, if the reviewer has specific suggestion regarding rearrangement of any particular figure, we will be happy to consider that.

*(3) Technical clarifications needed:*

*a) Specify exact concentrations used for inhibitors.*

We apologise for this unwanted and unnecessary mistake. Please note we will clearly mention the concentration of all the inhibitors used in this study in Result section and in Revised Figure legends. The revised section will read as, “Finally, we infected the KCs with GFP expressing LD-R for 4Hrs, washed and allowed the infection to proceed in presence of fluorescent red-LDL and Latrunculin-A ( 5µM), a compound which specifically inhibits fluid phase endocytosis by inducing actin depolymerization [41]. Real-time fluorescence tracking demonstrated that Latrunculin-A treatment not only prevented the uptake of fluorescent red-

LDL but also severely impacted intracellular proliferation of LD-R amastigotes (Movie 2A and 2B and Figure 4E). In contrast, treatment with Cytochalasin-D (2.5µg/ml), which alters cellular F-actin organization but does not affect fluid phase endocytosis”

*b) Include more details on image analysis methods.*

Please note that in specific sections like in Line numbers 574-579, 653-658, 1047-1049 of the Submitted manuscript, we have put special attention in describing the Image analysis process. However, we agree that in some particular cases more details will be appreciated by the reader. Hence we will be including an additional section of Image Analysis in the Methods section of the revised manuscript. This section will read as, “Image processing and analysis were conducted using Fiji (ImageJ). For optimal visualization, Giemsa-stained macrophages (*MΦs*) were represented in grayscale to enhance contrast and structural clarity. To improve the distinction of different fluorescent signals, pseudo-colors were assigned to fluorescence images, ensuring better differentiation between various cellular components. For colocalization analysis (Figures 3, 5, 6, and S2), we utilized the RGB profile plot plugin in ImageJ, which allows for the precise assessment of signal overlap by generating fluorescence intensity profiles across selected regions of interest. This approach provided quantitative insights into the spatial relationship between labeled molecules within infected cells. Additionally, for analyzing the distribution of cofilin in Figure 4, the ImageJ surface plot plugin was employed. This tool enabled three-dimensional visualization of fluorescence intensity variations, facilitating a more detailed examination of cofilin localization and its potential reorganization in response to infection.”

*c) Clarify statistical analysis procedures.*

Response: We have already provided a dedicated section of Statistical Analysis in the Methods section and also have also shown the groups being compared to determine the statistical analysis in the Figure and in the Figure Legends of the Submitted manuscript. Furthermore, we plan to add additional clarification regarding the statistical analysis performed Revised manuscript. For example, in the Revised manuscript this section will read as, “All statistical analyses were performed using GraphPad Prism 8 on raw datasets to ensure robust and reproducible results. For datasets involving comparisons across multiple conditions, one-way or two-way analysis of variance (ANOVA) was conducted, followed by Tukey’s post hoc test to assess pairwise differences while controlling for multiple comparisons. A 95% confidence interval (CI) was applied to determine the statistical reliability of the observed differences. For non-parametric comparisons across multiple groups, Wilcoxon rank-sum tests were employed, maintaining a 95% confidence interval, which is particularly useful for analysing skewed data distributions. In cases where only two groups were compared, Student’s t-test was used to determine statistical significance, ensuring an accurate assessment of mean differences. All quantitative data are represented as mean ± standard error of the mean (SEM) to illustrate variability within experimental replicates. Statistical significance was determined at  $P \leq 0.05$ . Notation for significance levels: \* $P \leq 0.05$ ; \*\* $P \leq 0.001$ ; \*\*\* $P \leq 0.0001$ .”

*(4) Minor corrections:*

*a) Methods section could benefit from more details on Raman spectroscopy analysis.*

We agree with this suggestion of the Reviewer. For providing more clarity we will incorporate additional details in the Methodology for the Raman section of the Revised manuscript. The updated section will read as follows in the revised manuscript. “For confocal Raman spectroscopy, spectral data were acquired from individual cells at 1000× magnification using a 100 × 100 µm scanning area, following previously established specifications. After spectral acquisition, distinct Raman shifts corresponding to specific biomolecular signatures were

extracted for further analysis. These included: Cholesterol ( $535\text{--}545\text{ cm}^{-1}$ ), Nuclear components ( $780\text{--}790\text{ cm}^{-1}$ ), Lipid structures ( $1262\text{--}1272\text{ cm}^{-1}$ ), Fatty acids ( $1436\text{--}1446\text{ cm}^{-1}$ ). Following spectral extraction, pseudo-color mapping was applied to highlight the spatial distribution of each biomolecular component within the cell. These processed spectral images are presented in Figure 3D1, where the first four panels illustrate the individual biomolecular distributions. A merged composite image was then generated to visualize the co-localization of these biomolecules within the cellular microenvironment, with the final panel specifically representing the spatial distribution of key biomolecules.”

*b) In the methods section line 609, page 14, the authors cite Real F protocol as reference 73 for PV enrichment. However, in the very next section on GC-MS analysis (lines 615-616, page 15), they state they have used reference 74 for PV enrichment. Can they explain why a discrepancy in PV isolation references this? Reference 74 does not mention anything related to PV isolation.*

We would like to sincerely apologise for this confusion which probably raised from our writing of this section. We would like to confirm that our PV isolation protocol is based on the published work of Real F protocol (reference 73). However, in the next section of the submitted manuscript, GC-MS analysis was described and that was performed based on protocol referenced in 74. In the Revised manuscript, we will avoid this confusion and made correction by putting the references in the proper places. Revised section will read as,

“GC-MS analysis of LD-S and LD-R-PV

Following a 24Hrs infection period, KCs were harvested, washed with phosphate-buffered saline (PBS), and pelleted. Subsequent to this, PV isolation was carried out using the previously described method [73]. The resulting parasitophorous vacuole (PV) pellet was processed for sterol isolation for GC\_MS analysis following a previously established protocol [74], with slight modification. Briefly, the PV pellet was resuspended in 20 ml of dichloromethane:methanol (2:1, vol/vol) and incubated at  $4^{\circ}\text{C}$  for 24hours. After centrifugation (11,000 g, 1 hour,  $4^{\circ}\text{C}$ ), the supernatant was checked through thin layer chromatography (TLC) and subsequently evaporated under vacuum. The residue and pellet were saponified with 30% potassium hydroxide (KOH) in methanol at  $80^{\circ}\text{C}$  for 2 hours. Sterols were extracted with n-hexane, evaporated, and dissolved in dichloromethane. A portion of the clear yellow sterol solution was treated with N, O-bis(trimethylsilyl)trifluoroacetamide (BSTFA) and heated at  $80^{\circ}\text{C}$  for 1 hour to form trimethylsilyl (TMS) ethers. Gas chromatography/mass spectrometry (GC/MS) analysis was performed using a Varian model 3400 chromatograph equipped with DB5 columns (methyl-phenylsiloxane ratio, 95/5; dimensions, 30 m by 0.25 mm). Helium was used as the gas carrier (1 ml/min). The column temperature was maintained at  $270^{\circ}\text{C}$ , with the injector and detector set at  $300^{\circ}\text{C}$ . A linear gradient from 150 to  $180^{\circ}\text{C}$  at  $10^{\circ}\text{C}/\text{min}$  was used for methyl esters, with MS conditions set at  $280^{\circ}\text{C}$ , 70 eV, and 2.2 kV.”

<https://doi.org/10.7554/eLife.102857.1.sa0>